
When the Most Disruptive Messages Are Safe to Prune in Aggregate: Functional Analysis of Communication Pruning in Two-Agent LLM Systems

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Abstract

1 In two-agent LLM solver-checker systems, we uncover a *perturbation-sensitivity*
2 *dissociation*: acknowledgments—the most individually disruptive message type—
3 can be safely pruned in aggregate. We establish this finding across two backbone
4 families (Qwen2.5-7B, Llama-3.1-8B), two reasoning domains, and multiple scales
5 (7B primary, with preliminary 14B and 72B probes), confirmed by 10-seed repli-
6 cation ($p < 0.001$). Three acknowledgment detectors—an oracle classifier, a
7 heuristic rule, and a learned DistilBERT model—all yield non-inferior pruned
8 accuracy under non-inferiority testing, demonstrating that pruning safety is robust
9 to detection method and precision.

10 Acknowledgment pruning eliminates 36–56% of inference calls (entire LLM invo-
11 cations, not token-level compression) while maintaining task accuracy on GSM8K
12 and StrategyQA. Correlational evidence is consistent with a *cascade dissolution*
13 pattern in which self-reinforcing acknowledgment loops dissolve without degrading
14 information flow. Boundary conditions—heuristic non-transferability, domain mis-
15 match, and stochastic decoding sensitivity—motivate empirical screening before
16 deployment. **All results are under greedy decoding in two-agent topologies.**

1 Introduction

18 LLM-based two-agent solver-checker systems communicate prolifically—a single math problem
19 can generate dozens of messages—yet which are structurally necessary remains open. Existing
20 approaches operate at different granularities: token-level compression Jiang et al. [2023], Chevalier
21 et al. [2023], runtime trajectory filtering Lin et al. [2026], agent-level pruning Wang et al. [2025c],
22 Zhang et al. [2025a], message-level suppression Li et al. [2025a], Zeng et al. [2025], and system-
23 level optimization Chen et al. [2025]. Yet none perform *functional analysis*—understanding what
24 communicative role each message plays—nor validate whether functional classification predicts
25 which messages can be safely removed.

26 An intuitive hypothesis is common: high-information messages are important; low-information
27 acknowledgments are expendable. This conflates *perturbation sensitivity*—how much removing a
28 message disrupts flow—with *informational importance*—whether it carries task-relevant content.
29 Pruning methods that score by content richness Wang et al. [2025c], Zeng et al. [2025] implicitly
30 assume these correlate. Yet conversation analysis shows backchannels Yngve [1970], Schegloff
31 [1982], Jefferson [1984] are structurally load-bearing despite negligible propositional content Clark
32 and Schaefer [1989]. Whether LLM agents exhibit analogous dependencies has never been tested.

33 **The gap.** Three layers of evidence point in contradictory directions: (1) *information content*
34 suggests acknowledgments are dispensable; (2) *individual perturbation* reveals they are structurally

critical (91% high disruption); (3) *aggregate pruning* shows they are safely removable ($\Delta \approx 0$ pp). No prior work has tested this three-way paradox in LLM agents nor explained why individually critical messages become safely removable in aggregate Rizvi-Martel et al. [2026].

Contributions. To our knowledge, we present the first perturbation-based empirical study of message-level pruning in two-agent solver-checker LLM systems, spanning two primary backbones (Qwen, Llama; Gemma exploratory), two validated task domains (with ARC and MATH as supplementary), and three topologies (15 configurations at 7B, with 14B and preliminary 72B scale probes):

1. **Perturbation-sensitivity dissociation** (§5.1–5.3). Messages most disruptive to perturb individually can be pruned in aggregate without significant accuracy loss. We conjecture that self-reinforcing ack cascades create structural redundancy that dissolves under aggregate removal (Qwen 90.7%, Llama 61.9%)—a mechanism distinct from human backchannel dynamics, requiring causal verification. This finding is enabled by a reusable analysis protocol comprising a 5-class taxonomy, structured annotation, and perturbation-based impact measurement.
2. **Analysis protocol** (§4.2–4.4). A 5-class communicative function taxonomy validated by cross-model agreement ($\kappa=0.721$) and human annotation ($\kappa=0.76$), a structured annotation protocol (constrained JSON, cross-model validation), and a perturbation-based impact measurement pipeline with non-inferiority and TOST equivalence testing—designed to be applicable to any two-agent LLM system for message-level communication analysis. The taxonomy serves a dual role: as a scientific lens for understanding communication structure, and as a label generator enabling automated pruning without per-message annotation at deployment.
3. **Pruning: from oracle to deployment** (§5.2, 5.5). Oracle and annotation-free heuristic pruning reduce inference calls by 36–56% with non-inferiority at $\delta = -3$ pp ($p < 0.001$) and supplementary Qwen-specific TOST equivalence at ± 1 pp (oracle $p = 0.048$, heuristic $p = 0.007$; 10 seeds each), validated across two backbone families and two task domains under greedy decoding. Boundary conditions—heuristic non-transferability (Qwen3), domain mismatch (MATH), and backbone-dependent stochastic decoding sensitivity—motivate empirical screening ($N \geq 200$) and a learned-classifier path beyond hand-crafted rules.

Scope. Claims are validated under greedy decoding; stochastic decoding ($t=0.7$) is validated for Qwen (10 seeds, TOST $p < 0.001$ at $\delta = -3$ pp) with Llama showing a marginal degradation trend (§5, §6). Our message-level analysis complements token-level compression Jiang et al. [2023] and agent-level pruning Wang et al. [2025c], bridging theoretical bounds Rizvi-Martel et al. [2026] and practical pruning. The taxonomy serves as an analytical lens; heuristic detectors that make it unnecessary at deployment are products of the analysis (§2).

2 Related Work

Our work draws on conversation analysis, multi-agent communication, and message function analysis.

Conversation analysis and grounding. Human conversation relies on low-content but structurally essential signals that coordinate mutual understanding. Backchannels Yngve [1970], continuers Schegloff [1982], and acknowledgment tokens Jefferson [1984] carry little propositional content yet are load-bearing in conversational grounding Clark and Schaefer [1989]; Bavelas et al. [2000] demonstrate that reducing listener backchannels degrades narrative quality, confirming their functional necessity. Computational extensions have addressed backchannel prediction Liermann et al. [2023], Inoue et al. [2025] and representation learning Qian and Skantze [2026], but exclusively in human conversation; whether analogous functional dynamics hold for LLM-generated acknowledgments remains untested.

Multi-agent communication. Research on multi-agent communication spans from structured protocols to emergent language, addressing different facets of *how*, *when*, and *what* agents should exchange. Early work defined formal message types through agent communication languages Finin et al. [1994], while learned communication Sukhbaatar et al. [2016], Foerster et al. [2016], Wang et al. [2020] optimized bandwidth end-to-end. Recent theoretical contributions address complementary dimensions: Rizvi-Martel et al. [2026] derive bounds on *when* communication helps, Zheng et al. [2025] explore *alternatives* such as latent thoughts, and Chen et

87 *al.* Chen et al. [2026b] identify “what to communicate” as a key open question. At the application
 88 level, multi-agent debate Du et al. [2024] and LLM orchestration frameworks such as CAMEL Li
 89 et al. [2023], MetaGPT Hong et al. [2024], AutoGen Wu et al. [2024], AgentVerse Chen et al. [2024],
 90 and MegaAgent Wang et al. [2025a] confirm that communication structure affects downstream
 91 performance, though debate may collapse to majority voting Choi et al. [2025] and intrinsic model
 92 strength can dominate architectural choices Wu et al. [2025], Kaushal and Singh [2026], La Malfa
 93 et al. [2025]; see Yan *et al.* Yan et al. [2025] and Tran *et al.* Tran et al. [2025] for surveys. Zhou *et*
 94 *al.* Zhou et al. [2025b] and Buscemi *et al.* Buscemi et al. [2025] investigate *why* agents communicate
 95 but neither performs message-level functional analysis. Our work fills this gap by studying which
 96 natural-language messages are safe to *remove* based on their communicative function.

97 **Communication efficiency.** Existing efficiency methods operate at different granularities, but
 98 none analyze the communicative *function* of individual messages. At the coarsest level, Agent-
 99 Dropout Wang et al. [2025c, 2026] prunes entire agents, achieving 20–56% token reduction without
 100 distinguishing which *messages* are dispensable. Finer-grained approaches target turns and edges:
 101 S²-MAD Zeng et al. [2025] applies turn-level early stopping, OPTIMA Chen et al. [2025] op-
 102 timizes topology and routing, and SafeSieve Zhang et al. [2025c] prunes inter-agent edges via
 103 dual-mechanism scoring. Lin *et al.* Lin et al. [2026] introduce SupervisorAgent, a lightweight
 104 meta-agent that uses LLM-free adaptive filtering at runtime to reduce token consumption by ~30%,
 105 though their filtering operates at the token/trajectory level rather than at the level of communicative
 106 function. Complementary directions include agent pruning Zhang et al. [2025a], Li et al. [2025a],
 107 Xiao et al. [2026], Wang and Tong [2025], latent-space communication Zou et al. [2025], Du et al.
 108 [2025b], KV-cache sharing Ye et al. [2025], and holistic optimization Li et al. [2025b], Zhang et al.
 109 [2025b], Jiang et al. [2026] (Appendix Table 34). In contrast, our taxonomy-driven pruning targets
 110 specific message categories identified through functional analysis, and we provide the first evidence
 111 that pruning safety is invariant across a $10\times$ parameter range (7B–72B).

112 **Topology and protocol.** A parallel thread investigates *who* communicates and *how*, optimizing
 113 organizational structure rather than message content. Automated topology design methods such as
 114 GoAgent Chen et al. [2026a] and AnyMAC Wang et al. [2025b] search over communication graphs,
 115 while LACP Li et al. [2025c] and GUARDIAN Zhou et al. [2025a] learn adaptive protocols. Shen *et*
 116 *al.* Shen et al. [2025] analyze how topology shapes information propagation. Benchmarks including
 117 ProtocolBench Du et al. [2025a] and MultiAgentBench Zhu et al. [2025] provide standardized
 118 evaluation for these structural choices. We complement this body of work by analyzing *what* is
 119 communicated at the message-function level, an orthogonal dimension that topology-focused methods
 120 leave unexamined.

121 **Attribution and pragmatics.** Attribution methods for multi-agent systems—including Hive-
 122 Mind Xia et al. [2026], AgentSHAP Horovicz [2025], and MAST Cemri et al. [2025]—assign
 123 credit at the agent or trace level, but do not decompose individual messages by communicative
 124 function; our ablation operates at this finer granularity. Our functional taxonomy draws on classical
 125 speech act theory Austin [1962], Searle [1969] and established dialogue act annotation schemes such
 126 as DAMSL Core and Allen [1997] and ISO 24617-2 Bunt et al. [2012]. Recent work has begun
 127 applying these frameworks to LLM settings: Qamar *et al.* Qamar et al. [2025] annotate dialogue acts
 128 in LLM conversations, Maitra *et al.* Maitra et al. [2025] use them as a lens on human–LLM interaction
 129 quality, and Zhang *et al.* Zhang et al. [2025b] find that acknowledgment and hints play key roles
 130 in dual-agent math problem-solving. We extend this line of inquiry to inter-agent communication,
 131 connecting functional categories directly to pruning outcomes.

132 3 Taxonomy of Communication Functions

133 We define five exhaustive, mutually exclusive communication function types (Table 1). Ambiguity
 134 is resolved by hierarchical priority: *coordination* $\succ$ *verification* $\succ$ *reasoning_guidance* $\succ$ *informa-*
 135 *tion_provision* $\succ$ *acknowledgment*, reflecting the principle that structurally higher-level functions
 136 take precedence.

137 Our categories map onto established dialogue act frameworks (DAMSL Core and Allen [1997], ISO
 138 24617-2 Bunt et al. [2012], FIPA ACL Foundation for Intelligent Physical Agents [2002]; detailed
 139 mapping in Appendix N), collapsing 56+ fine-grained types into five categories operationalized via

Table 1: Five-class taxonomy of communication functions in solver-checker dialogues, with definitions and GSM8K examples. Priority: coordination $\succ$ verification $\succ$ reasoning $\succ$ information $\succ$ acknowledgment.

Function Type	Definition	Example
INFORMATION PROVISION	Supplies factual content, data, or intermediate results	“The total cost is $3 \times \$5 = \15 for the shirts.”
REASONING GUIDANCE	Directs reasoning through decomposition or strategic suggestions	“First calculate the weekly cost, then multiply by 4.”
VERIFICATION	Checks or corrects the correctness of prior messages	“ $17 \times 3 = 51$, not 53. The corrected answer is \$51.”
COORDINATION	Manages dialogue flow, assigns roles, or delegates tasks	“I’ll solve first, then you verify step by step.”
ACKNOWLEDGMENT	Simple confirmations or agreements with no new content	“Your solution is correct.”

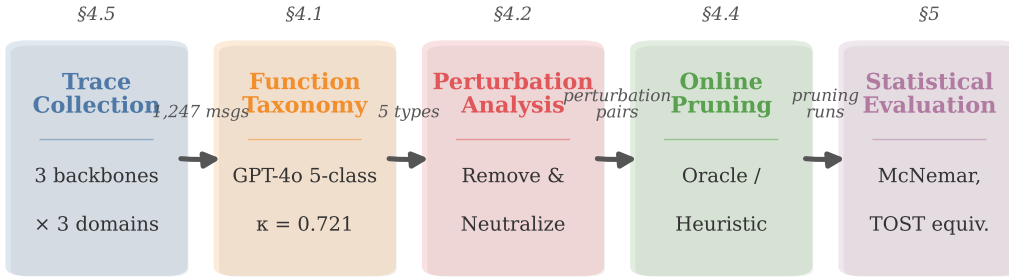


Figure 1: Experimental pipeline. Multi-agent conversations ($2+1$ backbones $\times 3$ domains, 15 configurations) are annotated into a 5-class taxonomy (cross-model $\kappa=0.721$), subjected to single-message perturbation analysis (remove and neutralize ablation), then pruned online under oracle or heuristic detection, and evaluated via McNemar exact tests and TOST equivalence.

measurable perturbation sensitivity rather than speaker intent. The taxonomy was developed for two-agent solver-checker dialogue; applicability to other configurations remains untested.

Each message is annotated by an LLM annotator using a structured prompt with taxonomy definitions and hierarchical decision rules (temperature 0, constrained JSON; Appendix K). Cross-model validation uses two independent annotator models per message ($\kappa \geq 0.4$ threshold; Section 5.4).

4 Experimental Setup

4.1 Analysis Protocol Overview

Existing MAS communication analyses operate at the turn level (which agent speaks when) or the system level (total token count). Neither captures the functional heterogeneity *within* turns: a single checker response may acknowledge, verify, and redirect in successive sentences. We therefore adopt message-level granularity, classifying each message by its communicative function rather than its surface form.

The 5-class taxonomy—acknowledgment, verification, reasoning guidance, information provision, coordination—is derived from speech act theory Searle [1969] and conversation analysis Schegloff [1982], Jefferson [1984], adapted to the structured interactions of LLM solver-checker systems. Five classes balance expressiveness against annotator reliability: coarser schemes (ack vs. non-ack) lose

the impact gradient; finer schemes degrade inter-annotator agreement below actionable thresholds. The analysis protocol is designed for reuse: researchers can apply the taxonomy schema, annotation protocol, and statistical evaluation pipeline to any LLM-based multi-agent system without redesigning the classification or assessment methodology.

4.2 Annotation Protocol

The annotation protocol is designed for reproducibility and portability across MAS configurations. Using the taxonomy from Section 3, each message is annotated via structured prompt with taxonomy definitions and hierarchical decision rules (temperature 0, constrained JSON output). To avoid self-annotation bias, messages generated by backbone A are annotated by backbone B . Non-blinding could inflate ack classifications but makes the pruning test more conservative. The protocol requires only a taxonomy-aware LLM annotator and constrained decoding—no task-specific training data—making it directly applicable to new backbone or domain configurations. Our core claims depend on the ack/non-ack binary distinction (Cohen’s $\kappa=0.721$ Cohen [1960], substantial agreement); fine-grained per-type agreement varies ($\kappa > 0.8$ for acknowledgment, $\kappa=0.237$ for information provision), so sub-category analyses are labeled preliminary. A human validation study ($N = 225$, $\kappa = 0.76$) is reported in Section 6.

4.3 Perturbation-Based Impact Measurement

To isolate each message’s contribution to downstream behavior, we introduce two complementary ablation procedures applied independently to every message in a recorded conversation trace.

Remove ablation. Message m_i is deleted from trace $\mathcal{T}$, and all subsequent messages are discarded and regenerated from position $i + 1$ onward. The behavioral impact is:

$$\text{sim}_{\text{remove}}(m_i) = \cos(\mathbf{e}(a_{\text{orig}}), \mathbf{e}(a_{\text{regen}})) \quad (1)$$

where a_{orig} denotes the final solver answer in the original trace and a_{regen} the answer regenerated from position $i+1$ after removing m_i ; $\mathbf{e}(\cdot)$ denotes the sentence embedding from all-MiniLM-L6-v2 Reimers and Gurevych [2019]. Cosine similarity measures perturbation sensitivity—behavioral disruption from message removal. This is distinct from pruning safety, which is independently evaluated via task accuracy and McNemar’s exact test (§4.5, Table 2). The impact gradient (§5.1) characterizes perturbation patterns; it does not serve as the basis for pruning safety claims.

Neutralize ablation. Message m_i is replaced with a type-matched placeholder that preserves structural position while removing informational content. Comparing remove and neutralize ablation disentangles a message’s *content* from its *structural presence*.

Each ablation is a single-message perturbation using greedy decoding via vLLM Kwon et al. [2023] on $4 \times$ NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (96 GB each), with conversations limited to $K = 6$ rounds.

Operational definitions. *Perturbation sensitivity*: $1 - \text{sim}_{\text{remove}}(m_i)$ (Eq. 1). *Informational importance*: task-relevant content operationalized via bigram novelty, token length, error correction rate, and Jaccard overlap.

4.4 Statistical Framework

The evaluation pipeline generalizes beyond acknowledgment pruning: any message-level intervention in a multi-agent system can be assessed by (1) measuring per-type perturbation sensitivity via single-message ablation, (2) testing aggregate pruning safety via McNemar’s exact test and non-inferiority bounds, and (3) quantifying equivalence via TOST. We instantiate this pipeline as follows.

We use OLS ANCOVA (Equation 2) with function type as the factor and position/round as covariates, plus linear mixed-effects (ME) models (Equation 3) with task as a random intercept (the maximal converging specification):

$$y_i = \beta_0 + \sum_k \beta_k \cdot \text{FT}_{ik} + \gamma_1 \cdot \text{Pos}_i + \gamma_2 \cdot \text{Round}_i + \epsilon_i \quad (2)$$

$$y_{ij} = \beta_0 + \sum_k \beta_k \cdot \text{FT}_{ijk} + \gamma_1 \cdot \text{Pos}_{ij} + \gamma_2 \cdot \text{Round}_{ij} + u_j + \epsilon_{ij} \quad (3)$$

where $u_j \sim \mathcal{N}(0, \sigma_u^2)$. Per-configuration ME models use individual tasks as random intercepts (maximal converging specification). Function type significance is tested via LRT with Benjamini–Hochberg correction ($\alpha = 0.05$; nine primary tests). End-to-end pruning safety is assessed primarily via confidence intervals on accuracy deltas (e.g., Qwen oracle 10-seed 95% CI $[-0.33, +1.13]$ pp) and McNemar’s exact test McNemar [1947]. Non-inferiority testing at $\delta = -3$ pp provides supplementary evidence: at baseline $\sim 90\%$, this margin represents a $\sim 30\%$ relative increase in error rate ($10\% \rightarrow 13\%$), calibrated against observed seed-to-seed variance ($\text{SD} \approx 1\text{--}2.4$ pp) to separate genuine degradation from sampling noise (Appendix S.2) Lakens [2017]. TOST equivalence at ± 1 pp further quantifies precision where applicable (Qwen oracle $p = 0.048$, marginal; interpret alongside the CI).

Multiple comparison framework. We organize statistical tests into three families to control false discovery risk while preserving power. *Family A* (primary confirmatory): nine LRT tests for function-type effects in mixed-effects models receive Benjamini–Hochberg correction at $\alpha = 0.05$. *Family B* (online cross-domain): McNemar tests across the backbone $\times$ domain pruning matrix receive BH correction within family. *Family C* (offline exploratory): single-run offline comparisons (e.g., FG vs. ack-only, $N=100$) are explicitly labeled exploratory and do not receive cross-family correction. This hierarchy follows the confirmatory/exploratory distinction recommended in pre-registration practice; all Family C p -values are reported for transparency but should not be interpreted as confirmatory evidence.

220 4.5 End-to-End Pruning Protocol

We use *pruning* to refer to three distinct operations: (1) *remove-one ablation*—removing a single message to measure local impact (§4.3); (2) *offline replay*—removing all messages of a given type and replaying the conversation; (3) *online suppression*—intercepting and withholding messages in real time during live agent interaction. Context disambiguates throughout. Messages are suppressed per strategy and the conversation suffix is regenerated from the first suppression point. Four strategies are compared (§5.2): Baseline, Ack-only, FG (lowest-impact function type), and Random (10 seeds). Each “seed” controls problem selection (greedy decoding is deterministic).

Oracle detection. Two oracle pipelines establish upper bounds with perfect function-type detection. The *offline* pipeline precomputes GPT-4o labels on baseline conversations; the *online* pipeline annotates each message in real time within the pruned trajectory, eliminating distribution-shift confounds at higher cost. GPT-4o vs. GPT-4o-mini validation ($\kappa = 0.822$; Appendix S.21) confirms a cost-effective substitute.

233 4.6 Heuristic Detector Development

Because backbone choice determines communication surface form (§5.4), we develop *model-adapted* heuristic detectors that require no LLM annotation—analogue to task-specific prompt engineering, where rules are tuned to the target system rather than designed for universal transfer.

Qwen detector. A character-length threshold (< 30) exploits Qwen’s narrow ack surface band (42.2% of checker messages below 30 chars, median 4 tokens; Appendix T), yielding $F1 = 0.845$ on GSM8K with $\sim 42\%$ suppression.

Llama detector. Llama’s verbose acks (mean 59.8 tokens, 65.6% bigram novelty; Appendix T) render length thresholds near-vacuous. A composite detector combines keyword matching, 4-gram repetition ratio (> 0.3), and a relaxed length ceiling (< 300 chars), yielding $\sim 75\%$ suppression with comparable pruning safety (§5.2).

Both detectors are task-specific proof-of-concept classifiers; limitations are evaluated in Section 5.5 and discussed in Section 6.

4.7 Learned Classification

To quantify the gap between hand-crafted heuristics and learned detectors, we train supervised classifiers on the binary ack/non-ack label derived from the 5-class oracle annotations. The training corpus comprises 6,069 messages spanning all three backbones (Llama, Qwen, Gemma) and four domains (GSM8K, StrategyQA, ARC, MATH). We split by `task_id` hash (70/20/10 train/val/test) to prevent messages from the same problem instance from appearing in both training and evaluation sets—a critical design choice, since messages within a task share surface cues that would inflate held-out scores under random splitting.

TF-IDF + logistic regression. Character n -grams ($n=2-5$, `char_wb`) and word n -grams ($n=1-2$) are extracted via TF-IDF and fed to a balanced logistic regression classifier. This lightweight pipeline achieves test-set $F1 = 0.859$ (macro-averaged, $N=526$), substantially outperforming the best heuristic detector ($F1 = 0.542$), confirming that learned features capture ack surface patterns that hand-crafted rules miss across backbone–domain combinations.

Fine-tuning DistilBERT Sanh et al. [2019] on the same split achieves test-set $F1 = 0.867$ (macro; ack-class 0.822), comparable to TF-IDF+LR, confirming the lightweight pipeline as a practical alternative.

The cross-backbone, cross-domain training set ensures that classifiers generalize beyond any single configuration. Importantly, oracle pruning safety (§5.2) does not depend on detector quality—it establishes an upper bound under perfect detection—so learned classifiers serve as practical deployment tools rather than prerequisites for the pruning result.

4.8 Cross-Domain Experimental Design

A 3×3 factorial design (two primary backbones + one exploratory $\times$ three domains: GSM8K Cobbe et al. [2021], ARC-Challenge Clark et al. [2018], StrategyQA Geva et al. [2021]) yields 15 configurations (GSM8K under RR/SEQ/SEL; ARC and StratQA under RR only).

4.9 Implementation and Evaluation

Models and infrastructure. Two primary backbones—Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct—are selected for their contrasting communication styles: terse acknowledgments (Qwen median 4 tokens) versus verbose ones (Llama mean 59.8 tokens; §5.1). Scale probes at 14B (Qwen2.5-14B) and 72B (Qwen2.5-72B) test generalization beyond the 7B regime; Gemma-2-9B-it serves as an exploratory boundary condition for heuristic transferability (§5.5). Each agent team comprises a solver and a checker served via vLLM v0.6.6 Kwon et al. [2023] with greedy decoding (temperature = 0, top- $p = 1.0$) on $4 \times$ NVIDIA RTX PRO 6000 GPUs (96 GB each), orchestrated through AG2 v0.6.1 ConversableAgent with task-specific system prompts and a 10-round limit. Under selector topology, a third agent routes turns via GroupChat with automatic speaker selection. Function-type labels are assigned by two independent LLM annotators (Qwen2.5-7B as Model A, Llama-3.1-8B as Model B) with structured JSON output; oracle ablation and cross-model validation use GPT-4o (gpt-4o-2024-08-06). The 3×3 factorial design (§4.8) yields 15 configurations; combined with oracle, heuristic, and learned pruning strategies across multiple seed counts, the study spans approximately 120 GPU-hours.

Datasets and correctness criterion. Primary evaluation uses GSM8K Cobbe et al. [2021] (grade-school math, $N=200-1,000$ per configuration) and StrategyQA Geva et al. [2021] (commonsense reasoning, $N=200-238$). ARC-Challenge Clark et al. [2018] ($N=250$) serves as a cross-domain probe; MATH Hendrycks et al. [2021] ($N=200$, 102 valid after filtering) provides a boundary-condition test. Solution correctness is evaluated by exact match: extracted numerical answer for GSM8K, yes/no for StrategyQA, and letter for ARC-Challenge.

Evaluation metrics. Pruning effectiveness is quantified along two axes:

- *Accuracy delta:* $\Delta = \text{acc}_{\text{pruned}} - \text{acc}_{\text{baseline}}$, in percentage points (pp). Pruning is deemed *safe* under the non-inferiority criterion $\Delta \geq -3$ pp (§4.4).
- *Suppress rate:* $r_{\text{sup}} = n_{\text{suppressed}} / n_{\text{total}}$, the fraction of checker messages detected as acknowledgment and withheld during online pruning.

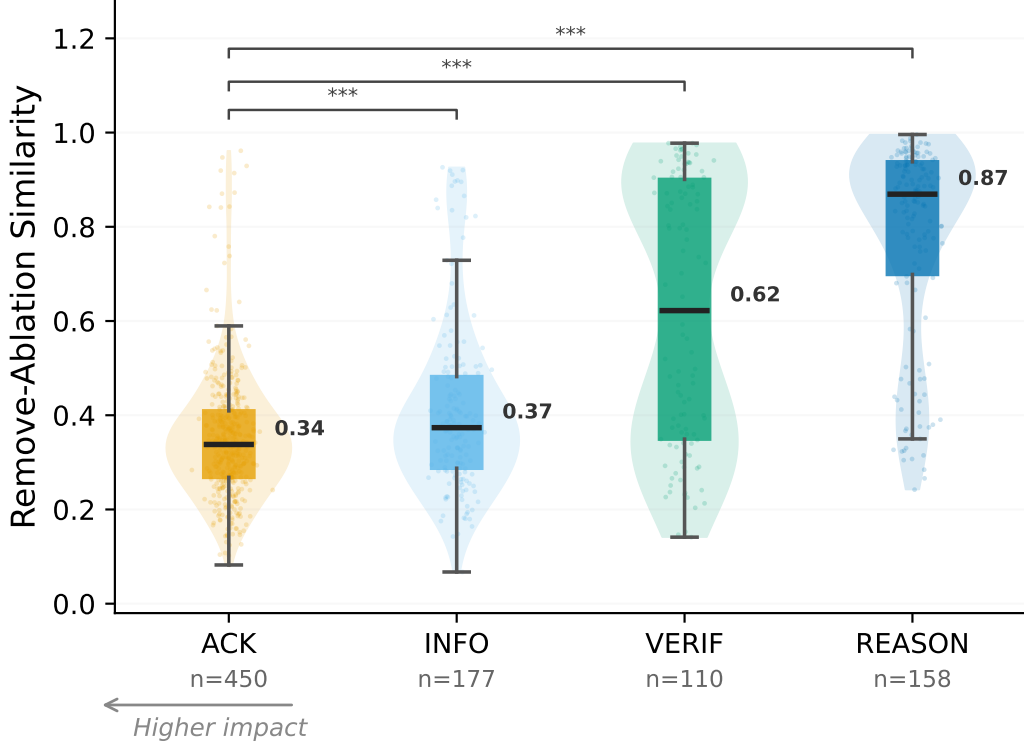


Figure 2: Remove-ablation similarity by function type (Qwen2.5-7B, round-robin, $n = 895$). Lower = greater disruption. Acknowledgments (median = 0.34) are most impactful despite least information content.

For single-seed configurations, statistical significance is assessed via McNemar’s exact test McNemar [1947] on the 2×2 concordance table of per-problem correct/incorrect outcomes between baseline and pruned runs (§4.4). Multi-seed configurations additionally report paired t -tests and 95% confidence intervals on seed-level Δ values, enabling TOST equivalence testing Lakens [2017].

Replication strategy. Each “seed” controls problem sampling order; generation is deterministic under greedy decoding. The flagship configuration (Qwen oracle, GSM8K) uses 10-seed replication ($10 \times 200 = 2,000$ task instances) to support tight confidence intervals and TOST equivalence testing. Secondary configurations (Llama GSM8K, both backbones on StrategyQA) use 5–8 seeds. Scale probes (72B) and boundary-condition tests (MATH, temperature = 0.7) are single- or few-seed and explicitly labeled exploratory. Code and data are available in the anonymous repository at [anonymized].

5 Results

We report results across 15 two-agent backbone $\times$ domain $\times$ topology configurations. The central question: in two-agent solver-checker systems, does perturbation sensitivity predict pruning safety?

5.1 Impact Gradient

For Qwen under fixed-turn topologies (Figure 2), the impact gradient is inversely related to information content: ACKNOWLEDGMENT (mean similarity = 0.352, $n = 450$), INFORMATION_PROVISION (0.410, $n = 177$), VERIFICATION (0.616, $n = 110$), REASONING_GUIDANCE (0.778, $n = 158$). Informationally sparse acknowledgments are those whose removal is most disruptive.

Pruning evidence. Removing a single acknowledgment is associated with high disruption (similarity < 0.5) in 91% of cases vs. 19% for reasoning guidance. The neutralize ablation produces the *opposite* gradient: neutralized information provision is associated with maximal disruption (0.047) while neutralized acknowledgments are associated with only moderate disruption (0.153; Appendix G). This double dissociation confirms acknowledgment impact stems from structural presence, not content. Robustness: placeholder choice $\rho = 0.73$; non-ack $\rho=0.90$; four similarity metrics $\rho > 0.91$ (Appendix I); RR/SEQ consistent, selector breaks down.

Taxonomy-independent information measures. Independent of taxonomic labels, acknowledgments are quantifiably information-sparse: Qwen acks have 5.7% bigram novelty (vs. 74.9% for reasoning guidance), median 4 tokens (vs. 117), and Jaccard overlap 0.69. Llama acks are longer (mean 59.8 tokens, 65.6% bigram novelty) but exhibit 0% error correction yet 14.3% error injection—a one-directional error channel (Appendix T).

5.2 E2E Communication Pruning

The individual perturbation results above suggest acknowledgments should be critical to preserve. The aggregate pruning tests below reveal the opposite—a dissociation we analyze in Section 6. We compare four strategies: **Baseline**, **Ack-only** (remove acknowledgments), **Function-Guided (FG)** (remove lowest-impact function type), and **Random** (same count as ack-only, 10 seeds).

Main results. Despite their individual disruptive impact, acknowledgments can be safely pruned in aggregate (Table 2, Figure 3). Both backbones show non-inferiority at $\delta = -3$ pp ($p < 0.001$). Ten-seed Qwen oracle: $\Delta = +0.40 \pm 1.02$ pp (95% CI $[-0.33, +1.13]$ pp; TOST $p = 0.048$ at ± 1 pp; 56.0% suppression). Ten-seed Llama oracle: $\Delta = +2.1 \pm 2.4$ pp (95% CI $[+0.4, +3.8]$ pp; 9/10 positive). The positive direction reflects a one-directional error channel (0/1,483 corrections but non-zero injection; Fisher $p = 0.004$). Non-inferiority ($p < 0.001$) provides the primary safety evidence; TOST provides supplementary equivalence precision (Qwen oracle $p = 0.048$, marginal; interpret alongside CI).

Message-level suppression (36–56%) translates to lower token-level savings ($\sim 10\%$ Qwen, $\sim 28\%$ Llama; Appendix W) because acknowledgments are shorter than substantive messages; the primary efficiency gain is inference-call elimination, not token reduction.

Heuristic detectors confirm deployment feasibility: Qwen 10 runs $\Delta = -0.05 \pm 0.99$ pp (TOST $p = 0.007$ at ± 1 pp). Llama 5 seeds $\Delta = +2.20 \pm 1.44$ pp (95% CI $[+0.41, +3.99]$ pp; non-inferiority $p < 0.001$); pruning is not merely non-inferior but suggestive of a small beneficial effect, though the interval width and limited seed count (5) preclude a strong improvement claim. The heuristic over-suppresses ($\sim 75\%$ vs. $\sim 37\%$ oracle rate), yet the positive direction suggests over-suppressed messages are predominantly redundant. Qwen StrategyQA (5 seeds) yields $\Delta = +0.70 \pm 0.91$ pp (95% CI $[-0.43, +1.83]$), consistent with non-inferiority ($p < 0.001$ at $\delta = -3$ pp). The Llama StrategyQA heuristic (8 seeds) yields $\Delta = +2.81 \pm 3.94$ pp (95% CI $[-0.49, +6.11]$), directionally positive but with high seed-level variance; non-inferiority holds (CI lower bound > -3 pp) with suppress rate stable across seeds (49–52%). Under stochastic decoding ($t=0.7$), pruning safety is backbone-dependent: Qwen oracle ($\Delta = +0.30 \pm 1.80$ pp, 10 seeds; TOST $p < 0.001$ at $\delta = -3$ pp) remains safe, while Llama shows marginal degradation (pooled $\Delta = -2.2$ pp, Fisher $p = 0.29$, NS; Appendix C).

In exploratory offline comparison ($N=100$, Qwen), FG pruning showed a substantial accuracy gap relative to ack-only (-5.0 pp vs. $+1.0$ pp, $p = 0.031$, exploratory), indicating aggregate safety is ack-specific. Online FG is confounded by cascading suppression ($\sim 90\%$ removal). Robust to 20% label noise (Appendix S).

Token-matched random baseline. Online random ($N=500$, Qwen) matches ack-only ($\Delta = +0.8$ pp, NS), but ack-only suppresses $5.7\times$ more messages (56.3% vs. 9.9%) without per-task calibration. Per-task paired comparison confirms near-identical outcomes: only 4/500 tasks differ between strategies (McNemar $p = 1.0$; ack risk of correct $\rightarrow$ wrong flip: 0.7%, identical to random),

[†]TOST ± 1 pp (marginal). [‡]Non-inferiority $\delta = -3$ pp. [§]Context overflow. [¶] $N=1,939$ after filtering ($\sim 3\%$ loss).
*Exploratory; confounded: suppress rate (79.3%) $\gg$ oracle ack rate ($< 5\%$); result reflects message redundancy, not ack-specific pruning.

Table 2: Online suppression pruning accuracy (%), N tasks, McNemar exact p). All rows use online suppression (messages intercepted during live agent interaction; cf. remove-one ablation in §4.3, offline replay in Appendix S). *Oracle*: GPT-4o 5-class ack-only. *Heuristic*: annotation-free rules ($F1 = 0.845/0.805$). Multi-seed rows report $\Delta \pm SD$ (pp). Supp. = messages suppressed.¹

Setting	N	Baseline	Pruned	p	Supp.
<i>Oracle (GPT-4o 5-class, ack-only)</i>					
Llama / GSM8K	500	82.2	81.6 (−0.6)	.710	36.6%
Qwen / GSM8K	500	88.8	89.6 (+0.8)	.340	56.3%
Llama / GSM8K ($t=0.7$)	500	82.2	79.2 (−3.0)	.150	28.0%
Qwen / GSM8K ($t=0.7$)	500	87.4	88.0 (+0.6)	.710	53.1%
Qwen / GSM8K ($t=0.7$, 10 seeds)	10×200	$\Delta = +0.30 \pm 1.80$ pp		<.001 [‡]	39.2%
Qwen / StratQA	500	73.2	73.4 (+0.2)	1.000	65.6%
Llama / StratQA [§]	238	68.9	73.9 (+5.0)	.180	21.5%
Qwen / ARC	250	90.8	91.6 (+0.8)	.500	14.6%
Qwen / GSM8K (10 seeds)	10×200	$\Delta = +0.40 \pm 1.02$ pp		.048 [‡]	56.0%
Llama / GSM8K (10 seeds)	10×200 [¶]	$\Delta = +2.1 \pm 2.4$ pp		<.001 [‡]	36.5%
<i>Token-matched random (online control)</i>					
Qwen / GSM8K	500	88.8	89.6 (+0.8)	.340	9.9%
<i>Heuristic (annotation-free, exploratory)</i>					
Llama / GSM8K (5 seeds)	5×200	$\Delta = +2.20 \pm 1.44$ pp		<.001 [‡]	74.9%
Qwen / GSM8K (10 runs)	10×200	$\Delta = -0.05 \pm 0.99$ pp		.880	41.8%
Qwen / StratQA (5 seeds)	5×200	$\Delta = +0.70 \pm 0.91$ pp		<.001 [‡]	56.9%
Llama / StratQA (8 seeds)	8×200	$\Delta = +2.81 \pm 3.94$ pp		.002 [‡]	50.3%
Qwen / GSM8K ($t=0.7$)	500	88.6	88.0 (−0.6)	.730	37.3%
Llama / GSM8K ($t=0.7$)	500	81.2	77.8 (−3.4)	.100	59.3%
Gemma / GSM8K*	300	75.3	76.0 (+0.7)	.683	79.3%
<i>Length-only baseline (char < 30)</i>					
Llama / GSM8K (5 seeds)	5×200	$\Delta = -0.10 \pm 1.29$ pp		—	0.8%
Qwen / GSM8K (3 seeds)	3×200	$\Delta = +0.50 \pm 1.50$ pp		—	42.2%
<i>Learned classifier (DistilBERT, $F1 = 0.867$)</i>					
Qwen / GSM8K	300	89.3	90.0 (+0.7)	.727	44.8%

364 demonstrating that acknowledgments are genuinely information-free—removing $5.7\times$ more of them
365 produces zero additional harm. For Llama, ack-only significantly outperforms matched-count random
366 removal (83% vs. 73.2%), ruling out conversation shortening as the explanation.

367 **Length-only baseline.** A character-length filter (< 30 chars) yields backbone-divergent results
368 (Table 2). For Qwen, it matches composite heuristic performance ($\Delta = +0.50 \pm 1.50$ pp, 3 seeds;
369 $p = 0.59$), confirming length as the primary discriminative axis. For Llama, the filter is near-vacuous
370 (0.8% suppression) while the composite heuristic suppresses $\sim 75\%$, confirming that keyword and
371 repetition rules are essential (§5.3).

372 **Learned classifier online validation.** Replacing heuristic rules with a fine-tuned DistilBERT
373 classifier (§4.7; $F1 = 0.867$) in the online pruning loop yields $\Delta = +0.7$ pp (89.3% $\rightarrow$ 90.0%,
374 $N=300$, McNemar $p = .727$, NS; 44.8% suppression; Table 2). Under our non-inferiority framing,
375 this non-significant result is the expected outcome: the McNemar test detects no accuracy difference
376 between pruned and baseline conditions, consistent with the NS pattern across all single-seed oracle
377 and heuristic configurations (Table 2). The discordant pair profile ($b = 3$, $c = 5$) reinforces
378 this interpretation: correct-to-wrong flips are rare (1.0%) and nearly balanced by wrong-to-correct
379 gains, indicating no directional harm from classifier-guided pruning. That pruning safety holds
380 across three detector types—heuristic rules ($F1 = 0.805\text{--}0.845$), learned classifier ($F1 = 0.867$),
381 and GPT-4o oracle—suggests safety depends on the pruning *target* (acknowledgments) rather than
382 detector precision. Compared to the Qwen heuristic ($\Delta = -0.05$ pp, 41.8% suppression), the learned
383 classifier achieves slightly higher suppression (44.8% vs. 41.8%) while maintaining equivalent
384 accuracy, demonstrating a practical drop-in replacement for annotation-free rules. DistilBERT
385 inference adds ~ 10 ms per message on CPU, negligible relative to vLLM generation latency.

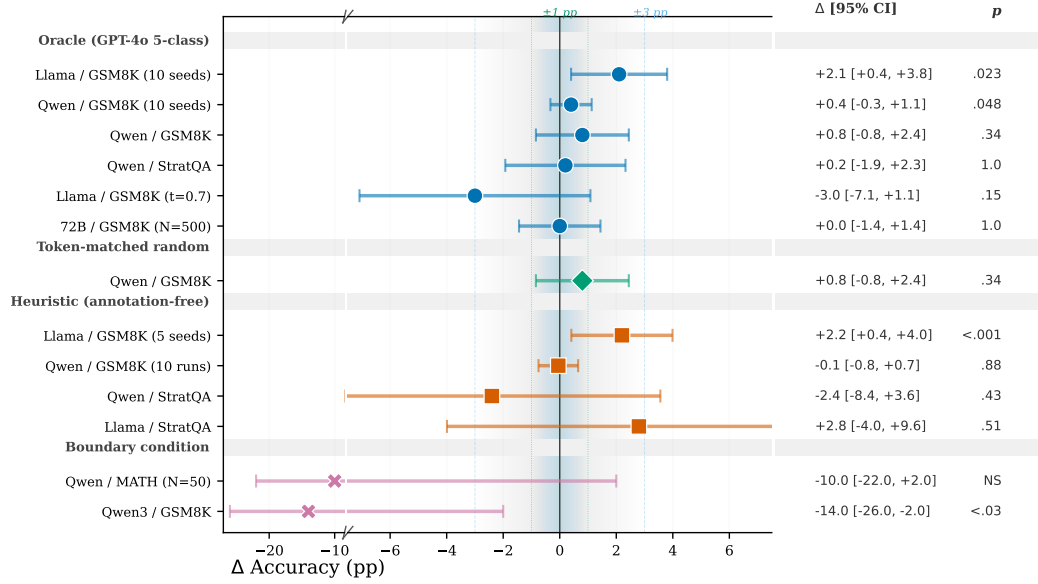


Figure 3: Forest plot of pruning accuracy deltas (Δ pp) with 95% CIs. Green/blue bands: ± 1 pp TOST / ± 3 pp practical margins. Oracle and heuristic multi-seed results cluster near zero for Qwen; Llama heuristic shows a positive trend. Boundary conditions (MATH, Qwen3) show degradation.

386 **Scale and heuristic detectors.** Pruning safety is directionally consistent across Qwen2.5 scales
387 (7B/14B/72B; Appendix U). Full-scale 72B oracle ($N=265$, single seed): $\Delta = +0.0$ pp ($p = 1.000$).
388 Heuristic detectors achieve $F1 = 0.845$ (Qwen) and 0.805 (Llama) without LLM annotation.

389 5.3 Backbone Communication Typology

390 Mixed-effects models ($N = 2,294$) reveal three topology-stable communication profiles: **function-**
391 **organized** (Qwen), **position-organized** (Gemma; exploratory— $\kappa = 0.35$, below threshold, $< 5\%$
392 **acks on GSM8K**), and **undifferentiated** (Llama). Backbone $\times$ function_type interaction is highly
393 significant ($p < 10^{-41}$), stable across topologies (three-way $p = 0.963$). Selector topology attenuates
394 Qwen’s function-organization and slightly amplifies Llama’s (Appendix S).

395 5.4 Cross-Domain Consistency

396 Backbone dominates communication patterns (Cramér’s $V = 0.45$ Cramér [1946] vs. 0.18 for
397 domain; Table 20). The 2×2 cross-backbone $\times$ cross-domain oracle matrix yields all cells NS
398 (Table 2), attributing any heuristic StratQA residual to detector error ($F1 = 0.21$); MATH defines a
399 domain boundary (§5.5).

400 **ARC-Challenge.** Oracle pruning on ARC-Challenge (Qwen, online re-generation; $N=250$, $\Delta =$
401 $+0.8$ pp, $p = .500$, 14.6% suppression) extends cross-domain validation to a third reasoning domain.
402 Offline simulation on Llama is directionally consistent with the online pattern: baseline $72.8\% \rightarrow$
403 $\text{pruned } 76.8\%$ ($N=250$, $\Delta = +4.0$ pp, $p = .013$, 37.1% suppression), strengthening dual-backbone
404 evidence on ARC despite the methodological difference (simulate mode re-extracts answers from
405 pruned traces rather than re-generating responses).

406 5.5 Boundary Conditions

407 Three boundary conditions define pruning limits. (1) *Domain mismatch*: MATH yields $\Delta = -2.0$ pp
408 (NS, $N=200$, 102 valid; Appendix J), with acks qualitatively different from GSM8K (16.6 tokens,
409 33% bigram novelty vs. 5.6 tokens, 0%). (2) *Heuristic non-transferability*: Qwen3-8B’s ack format
410 (25–59 chars) bypasses the v1 heuristic; a wider char <80 threshold reaches $\Delta = -14$ pp ($p < .03$),
411 confounded with non-ack suppression (Appendix Q). (3) *Online cross-domain*: Qwen StratQA

412 heuristic (5 seeds) yields $\Delta = +0.70$ pp (NS), consistent with non-inferiority; the earlier single-seed
 413 estimate (-2.4 pp, $N=250$) is superseded by the more reliable 5-seed replication ($N=200 \times 5$; oracle
 414 $+0.2$ pp). We recommend empirical screening ($N \geq 200$, $\geq 80\%$ power for ≥ 5 pp effects) before
 415 deployment.

416 6 Discussion

417 **Perturbation–sensitivity dissociation and cascade dissolution.** We hypothesize that the disso-
 418 ciation reflects LLM-specific cascade dynamics: each acknowledgment alters shared history via
 419 autoregressive conditioning, raising the next agent’s ack probability (Qwen 90.7%, Llama 61.9%,
 420 both $p < 10^{-13}$; Appendix S.13). Aggregate removal appears to dissolve these chains rather than
 421 accumulating errors, consistent with flat dose-response curves (Appendix S.13). Position-controlled
 422 logistic regression confirms genuine sequential dependence (previous-ack OR = 6.43 Qwen, 2.02
 423 Llama, both $p < 10^{-5}$; Appendix S.13). Notably, the position effect itself reverses across backbones
 424 (Qwen OR = 1.95, amplifying; Llama OR = 0.73, dampening), yet sequential dependence remains
 425 significant in both—ruling out a pure positional confound. The effect is ack-specific (FG degrades
 426 accuracy, $p = 0.031$, exploratory) and distinct from human backchannels Bavelas et al. [2000]; the
 427 double dissociation (§5.1) argues against a noise hypothesis. We conjecture that aggregate safety
 428 holds when: (i) cascade rate $P(\text{ack} | \text{ack}) > 0.5$ and (ii) bigram novelty below 10%. Qwen satisfies
 429 both (90.7%, 5.7%); Llama satisfies (i) but not (ii), with safety consistent with error-channel removal
 430 (Appendix S.17). Both conditions are violated in MATH where pruning degrades accuracy. Causal
 431 verification through controlled intervention remains needed; we emphasize that cascade dissolution
 432 is a descriptive framework for the observed correlational patterns, not a confirmed causal mecha-
 433 nism. These thresholds are post-hoc observations derived from two backbone families; prospective
 434 validation on additional architectures is needed before they can serve as deployment guidelines.

435 **Dual-mechanism robustness.** The pruning safety result is robust across two mechanistically
 436 distinct pathways. For Qwen, where acknowledgments are terse (median 4 tokens, 5.7% bigram
 437 novelty), pruning removes redundant structural signals—a *redundancy removal* mechanism. For
 438 Llama, where acknowledgments are verbose (mean 59.8 tokens, 65.6% bigram novelty) yet carry
 439 zero corrections and non-zero error injection (Fisher $p = 0.004$), pruning eliminates a one-directional
 440 *error channel*. That the dissociation holds across both mechanisms—arising from redundancy
 441 removal in one backbone and error-channel removal in the other—suggests a structural property of
 442 acknowledgment cascades rather than a mechanism-specific artifact. Exploratory Gemma heuristic
 443 pruning ($\Delta = +0.7$ pp, $N=300$, $p = .683$, NS; 79.3% suppression) is confounded: the heuristic
 444 suppresses 79.3% of messages despite $<5\%$ oracle ack rate (§5.3), indicating detection of short
 445 messages rather than acknowledgments. The positive Δ suggests high message redundancy in
 446 Gemma’s position-organized communication, but does not constitute ack-specific pruning evidence.
 447 This non-transferability reinforces that heuristic detectors require per-backbone calibration (§5.5).
 448 For Llama on GSM8K, the heuristic CI is entirely positive ($[+0.41, +3.99]$ pp, 5 seeds)—consistent
 449 with error-channel removal improving downstream signal quality—indicating that pruning is not
 450 merely safe but actively beneficial; on StrategyQA, heuristic pruning remains non-inferior (CI
 451 $[-0.49, +6.11]$ pp, 8 seeds) with higher seed-level variance, suggesting the magnitude of the effect
 452 is domain-modulated. Temperature robustness mirrors the typology distinction: Qwen’s function-
 453 organized cascade persists under stochastic decoding (92.2% at $t=0.7$ vs. 90.7% greedy), while
 454 Llama’s undifferentiated pattern is temperature-sensitive (suppress drops from 37% to 28%). This
 455 dual-mechanism evidence strengthens the case for the analysis protocol as the primary contribution:
 456 the protocol detects prunability regardless of the underlying mechanism.

457 **Prompt independence.** Removing the explicit confirmation instruction from the checker prompt
 458 leaves the heuristic ack detection rate at 76.2% (Qwen/GSM8K, $N=100$, $\text{char}<30$ heuristic), com-
 459 pared to 47.9% oracle ack rate under the standard prompt. The 76.2% figure reflects heuristic
 460 sensitivity to short messages, not oracle ack prevalence; the two are not directly comparable (different
 461 detection methods, different prompt conditions). Regardless, ack-like behavior persists without
 462 prompting, suggesting acknowledgments are emergent rather than prompt-driven. Primary pruning
 463 safety evidence (§5.2) is based entirely on oracle labels, not heuristic detection.

Communication structure and validation. Backbone dominates communication typology ($V = 0.45$ vs. 0.18 ; §5.4), with cross-model concordance confirming model-specific ack placement (stratified $\kappa = 0.018$; Appendix S.15). Suppression rates (36–56%) translate to equivalent inference-call reductions (Appendix W). Four-annotator validation confirms taxonomy reliability (pairwise $\kappa = 0.76$, 5-class; binary ack/non-ack $\kappa = 0.81$; Appendix S.18); Gemma’s $\kappa = 0.35$ falls below threshold and its results are treated as exploratory. Acknowledgment prevalence varies sharply with topology: debate produces near-zero acks (0.2%), and code-generation (HumanEval) only 4%, suggesting acknowledgments are an emergent property of solver-checker interaction rather than a universal feature of multi-agent communication.

Detector limitations. Heuristic detectors are proof-of-concept classifiers with non-transferability across domains (F1: $0.845 \rightarrow 0.213$), backbones (Gemma $\kappa = 0.35$), and generations (Qwen3-8B: –14 pp; §5.5). No universal length proxy exists ($56\times$ coverage gap). Oracle pruning remains robust across all tested cells, confirming the detector—not the pruning principle—is the limiting factor. An adversary who can inject acknowledgment-patterned messages into a multi-agent pipeline could exploit the cascade dynamics to cause outsized behavioral disruption with minimal detectable content; defenses should monitor structural patterns alongside content for integrity.

Limitations. Oracle labels face a distribution-shift caveat (offline vs. online gap: negligible for Qwen, larger for Llama; Appendix S.12). Pruning safety is backbone- and domain-specific (§5.5); all experiments use two-agent topologies under greedy decoding; scale evidence at 72B is preliminary (single seed). Additionally, annotation content previews for StrategyQA, ARC, and MATH are predominantly short (< 100 characters), so ack/non-ack binary labels—where acknowledgments are inherently brief—are unlikely affected, but fine-grained 5-class annotations on longer messages may suffer information loss from truncation. Online evaluation of the DistilBERT classifier (F1 = 0.867 ; §4.7) bridges the oracle-to-deployment gap on a single configuration (GSM8K, Qwen2.5-7B-Instruct, $N=300$, seed 42): $\Delta = +0.7$ pp (McNemar $p = .727$, NS; 44.8% suppression; §5.2), confirming no performance degradation; however, generalization across datasets, backbones, and random seeds remains unverified, and CPU inference latency (~ 10 ms/msg) may require GPU batching at production scale. Heuristic detectors remain non-transferable across domains (F1: $0.845 \rightarrow 0.213$). We recommend empirical screening ($N \geq 200$, $\geq 80\%$ power for ≥ 5 pp effects) before deployment; extended limitations appear in Appendix S.21.

7 Conclusion

Perturbation sensitivity and informational importance are dissociated in two-agent LLM systems: messages most disruptive to perturb individually can be safely pruned in aggregate. We establish this across two backbone families, two reasoning domains, and multiple pruning strategies, with 10-seed replication confirming non-inferiority at $\delta = -3$ pp ($p < 0.001$; human-annotation $\kappa = 0.76$; cross-model $\kappa = 0.72$). Three lines of correlational evidence are consistent with our *cascade dissolution* conjecture: self-reinforcing ack cascades (Qwen 90.7%, Llama 61.9%), near-zero cross-model concordance ($\kappa = 0.018$), and information-sparse acknowledgment profiles. For Llama, zero corrections among 1,483 pruned messages (Fisher $p = 0.004$) reveal a unidirectional error channel—acknowledgments carry errors but not corrections—explaining why pruning improves accuracy. This pattern is distinct from human backchannel dynamics and requires causal verification.

Practically, acknowledgment pruning eliminates 36–56% of inference calls (entire LLM invocations rather than token-level compression) without accuracy loss, replicated across GSM8K (oracle, 10-seed $\Delta = +0.40 \pm 1.02$ pp) and StrategyQA (Llama heuristic 8-seed $\Delta = +2.81 \pm 3.94$ pp, non-inferior at $\delta = -3$ pp, $p = .002$; Qwen oracle $\Delta = +0.2$ pp, $N=500$, NS). Preliminary single-seed evidence at 72B ($N=265$) suggests no degradation. **All experiments use two-agent topologies under greedy decoding**; heuristic detectors require per-backbone calibration.

Beyond the empirical findings, we contribute a reusable analysis protocol—5-class taxonomy, structured annotation, perturbation analysis, and non-inferiority/TOST evaluation—applicable independently or jointly to new backbone families, task domains, or agent topologies. Online evaluation of a learned DistilBERT classifier ($\Delta = +0.7$ pp, $N=300$, McNemar $p = .727$, NS; 44.8% suppression; §5.2) bridges the oracle-to-deployment gap: the non-significant McNemar result—consistent with the NS pattern across all oracle and heuristic configurations—confirms that lightweight learned detectors

517 can replace heuristic rules without accuracy loss. Future work should extend to multi-agent topologies
518 and additional backbone families.

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735 A Qualitative Examples of Communication Functions

736 Table 3 presents representative messages for each of the five communication function types identified
737 in our taxonomy, selected from annotations with full inter-annotator agreement and maximum
738 confidence scores. Examples are drawn from different backbone–topology combinations to illustrate
739 both the consistency of function definitions and the stylistic variations across models.

740 B HumanEval Cross-Domain Pilot

741 To provide preliminary evidence of cross-domain applicability, we ran a 50-task pilot using Qwen2.5-
742 7B with a hierarchical (solver–reviewer) topology on the HumanEval code generation benchmark
743 Chen et al. [2021].

²Despite its surface resemblance to ACKNOWLEDGMENT, both annotators classified this Llama message as REASONING GUIDANCE with full agreement. The ambiguity illustrates Llama’s minimally differentiated communication mode (Section 6): its messages are functionally interchangeable regardless of nominal category, making reliable annotation difficult for this backbone.

Table 3: Qualitative examples for each communication function type. All examples have inter-annotator agreement with confidence = 1.0. Messages are lightly edited for length.

Function Type	Configuration	Message Excerpt
REASONING GUIDANCE (2,327 agreed)	Qwen / RR	<i>“To solve this problem, we need to convert the length of the wire from feet to inches and then divide by the length of each piece. 1. Convert 4 feet to inches: Since 1 foot = 12 inches, 4 feet = $4 \times 12 = 48$ inches. 2. Divide the total length in inches by the length of each piece...”</i>
	Gemma / Selector	<i>“Here’s how to solve the problem step-by-step: Start: 2 apples. First throw: $2 + 5 = 7$ apples. Second throw: $7 - 1 + 6 = 12$ apples. The answer is 12.”</i>
	Llama / RR	<i>“I’m glad the solution was accurate. If you have another problem you’d like me to check, feel free to ask.”²</i>
VERIFICATION (1,001 agreed)	Qwen / RR	<i>“Step 1: Calculate the initial amount of money Johnny has. Johnny’s allowance: \$20. Extra money added: \$10. Total initial amount: $\\$20 + \\$10 = \\$30$. Step 2: Determine the amount after the investment triples. Tripling the initial amount: $\\$30 \times 3 = \\90. The answer is 90.”</i>
	Llama / RR	<i>“The given problem is solved step by step. The final answer is 36.”</i>
	Gemma / Selector	<i>“Here’s how to solve the problem: Find the total cost: \$12 (adult) + \$10 (child) = \$22. Add the change: $\\$22 + \\$8 = \\$30$. Answer: Brittany’s mom gave the cashier \$30.”</i>
ACKNOWLEDG- MENT (1,854 agreed)	Qwen / RR	<i>“That is correct. Each single bagel will cost \$1.00 after the discount. The answer is 1.”</i>
	Llama / RR	<i>“You’ve correctly summarized the steps and confirmed the calculations. The sequence of events and the final calculation are accurate. The answer is 43.”</i>
	Gemma / RR	<i>“Your solution is perfectly correct! You clearly laid out the steps and used algebra effectively to find the answer. The answer is 10.”</i>
INFORMATION PROVISION (471 agreed)	Gemma / RR	<i>“Here’s the solution: Sunday’s movies: They watch 4 movies / $2 = 2$ movies on Sunday. Total weekend movies: $4 + 2 = 6$ movies each weekend. Movies in 4 weeks: $6 \times 4 = 24$ movies. The answer is 24.”</i>
	Qwen / RR	<i>“To determine how much crepe paper James needs, we first calculate the perimeter of his rectangular backyard fence...”</i>
	Llama / RR	<i>“Here’s another problem: A bakery is making a special batch of cookies. They need to package 480 cookies into boxes that hold 12 cookies each. How many boxes can they fill with the 480 cookies? Let’s solve this problem step by step.”</i>
COORDINATION (595 agreed)	Llama / RR	<i>“I’m glad I was able to accurately solve the problem. If you have any more math problems you’d like me to solve, feel free to ask.”</i>
	Gemma / RR	<i>“Please provide me with the problem you’d like me to solve! I’m ready to check the solver’s solution...”</i>
	Llama / Selector	<i>“I’m glad you were able to provide the correct steps and solution. You have a good understanding of the problem...”</i>

Results. The system achieved 80% pass@1 accuracy (40/50 tasks). The communication function distribution differs markedly from GSM8K: REASONING_GUIDANCE dominates (50.8%), followed by VERIFICATION (41.5%), with ACKNOWLEDGMENT (4.2%), COORDINATION (1.7%), and INFORMATION_PROVISION (1.7%) appearing rarely. Cross-model annotation agreement reached $\kappa = 0.47$ (moderate), comparable to the lower end of the GSM8K range.

Limitations. The hierarchical topology proved too efficient for statistical analysis: 86% of tasks were solved in a single round (2 messages), yielding only 118 total messages and 18 eligible for local impact analysis. Most function types had $n < 30$, precluding Kruskal–Wallis and OLS ANCOVA tests. The distributional shift (from acknowledgment-heavy in GSM8K to reasoning-guidance-heavy in HumanEval) is itself informative—it suggests that the taxonomy captures domain-specific communication patterns—but the statistical framework requires substantially more multi-round conversations to produce meaningful results in this domain.

C Temperature Robustness

To assess sensitivity to decoding temperature, we ran a 50-task subset of Qwen $\times$ round-robin at temperature = 0.7 (primary experiments use greedy decoding). Accuracy remains stable (90.0% vs. 89.7%). The Kruskal–Wallis test remains significant ($H = 34.0$, $p < 0.001$), and OLS ANCOVA confirms function-driven communication ($F = 37.9$, $p < 0.001$; $R^2 = 0.50$; cross-model $\kappa = 0.60$). A parallel check on Llama $\times$ round-robin at temperature = 0.7 yields $R^2 = 0.10$, $\kappa = 0.41$, consistent with its minimally differentiated mode under greedy decoding. The function distribution shifts toward more REASONING_GUIDANCE (37% vs. 18%) and less ACKNOWLEDGMENT (36% vs. 50%), suggesting stochastic decoding produces more substantive content without altering the functional structure.

Pruning under stochastic decoding. All primary pruning results use greedy decoding. A 10-seed oracle replication on Qwen/GSM8K at $t=0.7$ (10×200 tasks) yields $\Delta = +0.30 \pm 1.80$ pp (95% CI $[-0.81, +1.41]$ pp; two-sided t : $p = 0.611$, NS; TOST non-inferiority at $\delta = -3$ pp: $p < 0.001$; at $\delta = -1$ pp: $p = 0.024$; 39.2% suppression). Per-seed deltas range from -3.0 to $+3.5$ pp: seed 6144 ($\Delta = -3.0$ pp) touches the non-inferiority boundary, while seed 1024 ($\Delta = +3.5$ pp) reflects an anomalously low baseline (85.0%, pruned 88.5%). The suppression rate (39.2%) is comparable to the greedy heuristic ($\sim 42\%$), indicating similar acknowledgment prevalence under stochastic decoding. In contrast, Llama oracle pruning at $t=0.7$ yields pooled $\Delta = -2.2$ pp (Fisher combined $p = 0.29$), an amplification of the marginal negative trend observed under greedy decoding ($\Delta = -0.6$ pp). Since both Qwen and Llama results use oracle labels, the divergence indicates that ack pruning safety under stochastic decoding is backbone-specific: Qwen remains consistently safe across 10 seeds, while Llama shows a marginal degradation trend ($p = 0.29$, NS) regardless of detector type. Temperature robustness cannot be assumed across backbones without empirical verification.

Gemma exhibits extreme temperature sensitivity: greedy ack suppression is 4.6% but rises to 63.6% under $t=0.7$ ($14\times$), suggesting Gemma’s position-organized communication pattern collapses under stochastic decoding. This reinforces Gemma’s exploratory status and indicates the three-backbone typology (§5.4) is greedy-specific.

D OLS Robustness Checks

We validate the OLS ANCOVA (Equation 2) with three robustness checks across all nine configurations.

Residual diagnostics. Shapiro–Wilk tests reject normality for all configurations (expected at $n > 400$); Breusch–Pagan tests detect heteroscedasticity in all cases. We therefore report HC3 robust standard errors throughout the main text. All conclusions are unchanged under robust inference. VIF analysis shows high collinearity between `message_position` and `round_number` ($VIF > 10$), but function type dummy VIF values remain below 3, confirming that multicollinearity does not affect the key `func_type` inference. Q-Q plots are available in the supplementary materials.

Nonlinear position model. We augment the linear model with a quadratic position term (position^2). Table 4 reports key comparisons. For Qwen and Llama, adding position^2 produces negligible $\Delta R^2 < 0.005$ and function type remains significant ($p < 0.001$). For Gemma, position^2 absorbs $\Delta R^2 = 0.228$ (RR) and 0.215 (SEQ), rendering function type non-significant ($p > 0.69$). This confirms that Gemma’s weak function-organization reflects genuine nonlinear positional dynamics: once position is properly modeled, function type adds negligible explanatory power.

Table 4: Linear vs. quadratic position model: function type significance after controlling for nonlinear position effects.

Configuration	Linear R^2	Quad R^2	ΔR^2	func_type p (quad)
Qwen RR	.451	.452	.001	< .001
Qwen SEQ	.443	.447	.004	< .001
Qwen SEL	.065	.065	.000	< .001
Gemma RR	.111	.339	.228	.806
Gemma SEQ	.123	.338	.215	.699
Gemma SEL	.010	.040	.031	.219
Llama RR	.021	.022	.001	.010
Llama SEQ	.019	.020	.001	.004
Llama SEL	.070	.074	.004	< .001

Message length control. Adding message length (character count) as a covariate produces $\Delta R^2 < 0.04$ for all configurations. Function type remains significant for Qwen ($p < 0.001$) and Llama ($p < 0.03$) after controlling for length, indicating that the impact ordering is not an artifact of message length differences across function types.

E Kruskal–Wallis Results

Table 5 reports Kruskal–Wallis H statistics testing whether function type predicts remove-ablation similarity across all nine backbone–topology configurations. All tests are significant ($p < 0.01$), confirming that the omnibus between-type differences reported in the main text (Section 5.1) are robust.

Table 5: Kruskal–Wallis H statistics for remove-ablation similarity across function types. All $p < 0.01$.

Backbone	RR	SEQ	Selector
Qwen2.5-7B	287.4	288.1	50.8
Gemma-2-9B	21.5	25.1	6.9
Llama-3.1-8B	13.0	15.8	63.8

F System Prompts

Full annotation system prompts are provided in Appendix K. The solver and checker system prompts used for multi-agent conversation are as follows:

Solver prompt. “You are a math problem solver. Solve the given problem step by step, showing all your work. Provide the final numerical answer clearly.”

Checker prompt. “You are a solution checker. Verify the solver’s solution step by step. If you find errors, point them out and provide the correct solution. If the solution is correct, confirm it.”

Selector prompt (selector topology only). “You are a group chat manager. Select the next speaker based on the current state of the conversation. Choose the solver when a new problem needs to be solved, and the checker when a solution needs to be verified.”

G Neutralize Ablation Analysis

While the main text reports remove-ablation impact, neutralize ablation (replacing a message with “Acknowledged.” while preserving its slot) measures a complementary dimension. The two ablation types produce *opposite* impact gradients:

- **Remove** (content + structure removed): acknowledgment most impactful (sim = 0.35), reasoning guidance least (0.78).
- **Neutralize** (content removed, structure preserved): information provision most impactful (sim = 0.05), verification least (0.50).

The difference $\Delta = \text{sim}_{\text{remove}} - \text{sim}_{\text{neutralize}}$ isolates the *structural value* of each message type. Information provision shows the largest $\Delta = 0.47$ (almost entirely content value), while coordination shows $\Delta = 0.07$ (almost entirely structural value). This decomposition is consistent with acknowledgment’s high remove-impact being driven by structural presence—violating conversational expectations—rather than informational content.

H Supplementary Tables

Table 6 reports the full cross-model annotation agreement matrix.

Table 6: Cross-model annotation agreement (Cohen’s κ) by backbone and topology.

Backbone	Round-Robin	Sequential	Selector
Qwen2.5-7B	0.64	0.64	0.669
Gemma-2-9B	0.350	0.348	0.351
Llama-3.1-8B	0.46	0.47	0.634

I Impact Metric Robustness

To verify that the impact gradient (Section 5.1) is not an artifact of the MiniLM embedding metric, we recompute message impact using three independent text similarity measures: ROUGE-L, token-level F1, and Jaccard similarity. Table 7 reports the Spearman rank correlation (ρ) between MiniLM cosine similarity and each alternative metric across all nine configurations.

Table 7: Spearman ρ between MiniLM cosine similarity and alternative impact metrics. All correlations computed over per-message impact scores within each configuration.

Configuration	ROUGE-L	Token F1	Jaccard
Qwen RR	1.0	1.0	1.0
Qwen SEQ	1.0	1.0	1.0
Qwen SEL	1.0	1.0	1.0
Gemma RR	1.0	1.0	1.0
Gemma SEQ	1.0	0.8	0.8
Gemma SEL	0.949	0.949	0.949
Llama RR	0.6	0.6	0.6
Llama SEQ	1.0	1.0	1.0
Llama SEL	0.9	0.9	0.9
Mean	0.939	0.917	0.917

The prevalence of $\rho = 1.0$ reflects a ceiling effect: with only 4–5 function types per configuration, Spearman rank correlation over group medians has limited resolution—any two metrics that preserve the same ordinal ranking produce $\rho = 1.0$ regardless of magnitude differences. Where rankings do diverge (Llama RR $\rho = 0.6$, Gemma SEQ $\rho = 0.8$), the disagreement involves adjacent categories with similar impact scores, not reversals of the overall gradient. The mean $\rho > 0.91$ across all metric pairs confirms that the impact gradient ordering is consistent across metrics, though the ceiling effect limits the strength of this claim for individual configurations.

J Cross-Domain Validation: MATH

To test whether the backbone communication typology generalizes beyond elementary arithmetic, we ran the full pipeline on 50 MATH tasks (Level 3–5, Algebra and Number Theory) using the same round-robin topology. A follow-up pruning replication at $N=200$ (102 valid after filtering tasks where both conditions failed) yielded $\Delta = -2.0$ pp (McNemar $p = 0.625$), attenuating the pilot’s -10 pp point estimate while remaining directionally negative (§5.5). The annotation analysis below is based on the original 50-task pilot.

Qwen on MATH. Accuracy is 72.0% (36/50), lower than GSM8K (89.7%) as expected for harder tasks. Each task produces 6 messages (300 total, 250 eligible for analysis after excluding initial prompts). Cross-model annotation agreement reaches $\kappa = 0.45$ (moderate, above threshold). The OLS ANCOVA yields $R^2 = 0.098$ with function type significant after position control ($F = 3.77$, $p = 0.011$); the mixed-effects model yields $R_m^2 = 0.201$ (LRT $p = 0.003$), confirming that Qwen’s function-driven mode persists on a harder mathematical domain, though with reduced effect size compared to GSM8K ($R_m^2 = 0.335$). The Kruskal–Wallis test is borderline ($H = 7.37$, $p = 0.061$), likely due to reduced sample size.

Gemma on MATH. Accuracy drops to 40.0% (20/50). Cross-model $\kappa = 0.48$ (above threshold). The OLS ANCOVA yields $R^2 = 0.236$ but function type is non-significant ($F = 0.09$, $p = 0.76$), while message position is highly significant ($F = 15.22$, $p < 0.001$). This replicates Gemma’s weak function-organization from GSM8K: on MATH, position again explains the variance while function type does not. The Kruskal–Wallis test is also non-significant ($H = 2.05$, $p = 0.15$).

Llama on MATH. Accuracy is 64.3% (27/42; 8 tasks produced empty traces). Cross-model $\kappa = 0.36$ (below threshold). With only 168 total messages, ACKNOWLEDGMENT dominates (56%), and most other function types fall below $n = 30$, precluding OLS ANCOVA and Kruskal–Wallis analysis. The acknowledgment-dominated distribution is directionally consistent with Llama’s minimally differentiated mode on GSM8K, though statistical confirmation requires larger samples.

Summary. Qwen’s function-organized structure ($F = 3.77$, $p = 0.011$) and Gemma’s position-organized structure ($F = 0.09$, n.s.; position $F = 15.22$, $p < 0.001$) both replicate on MATH. Llama’s distribution is directionally consistent with its undifferentiated profile but lacks statistical power. The backbone communication typology replicates beyond elementary arithmetic.

Why MATH acknowledgments carry task-critical content. MATH acknowledgments differ qualitatively from GSM8K acknowledgments, explaining the negative pruning direction (-2.0 pp at $N=200$, -10 pp in the underpowered $N=50$ pilot) under moderate functional organization ($R_m^2 = 0.201$). Of 93 messages receiving at least one acknowledgment label from dual annotators, 48 (51.6%) receive a non-acknowledgment label from the other annotator (typically INFORMATION_PROVISION), reflecting genuine boundary ambiguity. Representative examples illustrate the contrast:

- **MATH ack (annotator disagreement):** “The detailed step-by-step evaluation confirms that the number of integers m such that $0 < m \dots$ ” (Annotator A: INFORMATION_PROVISION; Annotator B: ACKNOWLEDGMENT). The message restates a multi-step derivation while confirming correctness.
- **MATH ack (both agree):** “The solution provided is correct. Here is the final answer boxed: 11” — confirms correctness but also provides the boxed answer that downstream agents parse.
- **GSM8K ack (both agree):** “The answer is 109.” (18 characters) — trivially redundant answer echo with no reasoning content.

Quantitative content analysis confirms the distinction: MATH acks average 16.6 tokens (vs. 5.6 for GSM8K), with median bigram novelty of 0.333 (vs. 0.0 for GSM8K, where acks introduce no new bigrams relative to the preceding message). Annotator disagreement on MATH acks reaches 55.9% (vs. 26.3% on GSM8K), reflecting genuine boundary ambiguity between acknowledgment and information_provision. Token overlap with the preceding message is similar (0.70 vs. 0.74), but the higher bigram novelty indicates that MATH acks frequently contain reasoning summaries—while confirmatory in intent, they carry solution steps that downstream agents rely on. Removing them eliminates not just structural anchoring but also task-critical content, violating the redundancy assumption that makes ack-only pruning safe on GSM8K.

897 K Annotation Prompt

898 The following system prompt is used for all LLM-based function type annotations. Each message
899 is annotated independently by two models (cross-model validation). Temperature is set to 0 with
900 constrained JSON output.

901 You are a communication analyst for multi-agent systems. Your task
902 is to classify the function of a message in a multi-agent math
903 problem-solving conversation.

904 Function Types

905 1. `information_provision`: Provides facts, data, intermediate
906 calculations, or partial results that advance the solution.
907 Includes reporting computed values, stating known quantities, or
908 presenting new numerical results.

909 2. `reasoning_guidance`: Provides reasoning direction, problem
910 decomposition, step-by-step plans, or conceptual frameworks.
911 Includes breaking the problem into sub-problems, suggesting solution
912 strategies, or outlining multi-step approaches.

913 3. `verification`: Checks, corrects, or confirms the correctness of
914 previous work. Includes identifying errors, validating results,
915 pointing out mistakes, or confirming a step is correct with a
916 specific reason.

917 4. `coordination`: Manages the flow of the conversation, assigns
918 tasks, requests clarification, or controls turn-taking. Includes
919 delegating sub-tasks, asking for more information, or directing the
920 problem-solving process.

921 5. `acknowledgment`: Simple confirmation, repetition, or agreement
922 that adds no new information or reasoning. Includes restating what
923 was already said without extending it.

924 Decision Rules

- 925 - If a message both provides new results AND checks previous work,
926 classify as `verification`.
- 927 - If a message provides a reasoning plan AND new results, classify as
928 `reasoning_guidance`.
- 929 - If a message confirms correctness WITH a substantive explanation,
930 classify as `verification`.
- 931 - If a message simply agrees or restates without adding new
932 information, classify as `acknowledgment`.

933 Output Format

934 Respond with ONLY a JSON object:
935 `{"type": "<function_type>", "confidence": <0.0-1.0>, "reasoning":`
936 `"<brief explanation>"}`

937 The user prompt provides the conversation context with all messages and marks the target message
938 for classification:

```
939 ## Conversation Context
940 [Message 0] {sender} -> {receiver}:
941 {content}
942 [Message 1] {sender} -> {receiver}: «< TARGET MESSAGE >>
943 {content}
944 [Message 2] {sender} -> {receiver}:
945 {content}
946 ...
947 Classify the function of Message {target_idx}.
```


L Mixed-Effects Model Results

To address within-trace non-independence (multiple messages per task), we fit linear mixed-effects models with task as a random intercept and the same fixed effects as the OLS ANCOVA (function type, normalized message position, round number). Both OLS and ME use remove-ablation similarity (Equation 1) as the dependent variable. Table 8 compares OLS and mixed-effects results. The likelihood ratio test (LRT) compares the full model against a reduced model without function type dummies.

Table 8: OLS ANCOVA vs. mixed-effects model (ME) with task random intercept. LRT = likelihood ratio test for function type. ICC = intraclass correlation from the full ME model. “Excluded” lists function types with $n < 30$ removed before fitting. *** $p < 0.001$; ** $p < 0.01$; n.s. = not significant. F and χ^2 values are not directly comparable across configurations due to differing numbers of included function types.

Config	n	OLS F	LRT χ^2	LRT p	ICC	Excluded
Qwen RR	895	239.6***	188.9***	<.001	.292	coord.
Qwen SEQ	894	230.2***	195.3***	<.001	.245	coord., unk.
Qwen SEL	891	58.2***	0.07 ^{n.s.}	.789	.716	r.g., i.p.
Gemma RR	883	7.82***	23.28***	<.001	.000	verif.
Gemma SEQ	882	9.04***	26.86***	<.001	.000	verif.
Gemma SEL	861	7.45**	3.66 ^{n.s.}	.056	.225	r.g., v., coord.
Llama RR	868	3.98**	16.31***	<.001	.297	coord.
Llama SEQ	870	4.83**	15.44**	.001	.205	coord.
Llama SEL	865	21.6***	36.89***	<.001	.126	i.p., r.g.

Key observations. (1) **Seven of nine configurations** show concordant OLS and ME conclusions: Qwen RR/SEQ remain strongly function-driven; Gemma RR/SEQ are significant in both OLS and ME but with small effect sizes ($R_m^2 = 0.11$ – 0.12); Llama remains significant across all topologies though with negligible effect sizes ($R_m^2 \leq 0.04$). (2) **Qwen SEL flips to non-significant** under ME (LRT $\chi^2 = 0.07$, $p = .789$). The ICC of 0.716 indicates that 72% of variance in remove-ablation similarity is between-task. With only two dominant function types (ACKNOWLEDGMENT and VERIFICATION) remaining after $n \geq 30$ filtering, the OLS effect reflects task-level confounding rather than within-task functional differentiation. (3) **Gemma SEL flips to non-significant** under ME (LRT $\chi^2 = 3.66$, $p = .056$, ICC = 0.23). With only two function types remaining after excluding three low-frequency categories, the borderline result has limited interpretability. (4) Both discrepancies occur under selector topology, which introduces strong task-level clustering. This suggests that dynamic routing creates task-dependent communication patterns that inflate OLS estimates.

M Mistral-7B Cross-Backbone Validation

To test whether the backbone communication typology generalizes beyond our three primary backbones, we ran the full pipeline on Mistral-7B-Instruct-v0.3 across all three topologies (300 GSM8K tasks each, same seed). Mistral uses the ChatML template with an explicit `<|im_end|>` stop token to prevent EOS residue in multi-turn conversations.

Accuracy. Mistral achieves 52% accuracy across all three topologies (RR: 52%, SEQ: 52%, SEL: 52%), substantially below Qwen (90%) and Gemma (80%) but comparable to Llama’s selector performance (56%).

Function type distribution. Mistral communication is dominated by coordination and acknowledgment (~90% combined), with minimal information provision (< 3%). Under selector topology, acknowledgment surges to 67%.

Communication mode: weakly differentiated. Table 10 summarizes the statistical analysis. Only the round-robin topology produces a significant function type effect (OLS $F = 6.6$, $p < .001$, $R^2 = 0.16$). Sequential and selector topologies show weaker or non-significant effects. Cross-model

Table 9: Mistral-7B function type distribution (%) by topology.

Function type	RR	SEQ	SEL
COORDINATION	55.5	50.7	25.4
ACKNOWLEDGMENT	34.9	41.7	66.9
REASONING_GUIDANCE	6.8	6.2	6.3
INFORMATION_PROVISION	2.7	1.4	0.7

981 annotation agreement is fair ($\kappa \approx 0.34$), below the primary backbones, consistent with Mistral’s less
 982 distinct functional boundaries.

Table 10: Mistral-7B statistical summary across topologies.

Topology	KW H	OLS F	R^2	κ
RR	22.4***	6.6***	.158	.344
SEQ	7.6	37.6***	.056	.353
SEL	5.2	1.5	.065	.315

983 **Interpretation.** Mistral occupies an intermediate position in the backbone communication typology,
 984 between Llama’s undifferentiated profile and Qwen’s function-organized structure. The structured
 985 turn-taking of round-robin activates a degree of functional differentiation that disappears under less
 986 constrained topologies. This *topology-sensitive* pattern—where communication structure depends on
 987 external routing rather than being a fixed model property—is consistent with the crossover interaction
 988 observed for the primary backbones.

989 **Implications for the typology.** Mistral’s intermediate position extends the backbone communica-
 990 tion typology beyond the three primary models. The four observed backbones—function-organized
 991 (Qwen, $R_m^2 = 0.34$), position-organized (Gemma, $R_m^2 = 0.11$), weakly differentiated (Mistral), and
 992 undifferentiated (Llama, $R_m^2 = 0.02$)—exhibit distinct communication structures, with topology
 993 acting as a modulating factor.

994 N DAMSL/ISO Taxonomy Mapping

995 Our five categories collapse 56+ fine-grained dialogue act types into operationally distinct categories
 996 defined by perturbation-based impact rather than speaker intent. Table 11 provides a detailed mapping
 997 to three established frameworks: DAMSL Core and Allen [1997], ISO 24617-2 Bunt et al. [2012],
 998 and FIPA ACL Foundation for Intelligent Physical Agents [2002].

999 **Design choices.** Several departures from standard dialogue act taxonomies reflect the LLM multi-
 1000 agent setting:

- 1001 • **Acknowledgment vs. backchannel.** Human backchannels (“uh huh,” “yeah”) are minimal tokens;
 1002 LLM acknowledgments range from terse echoes (Qwen: median 4 tokens) to verbose paraphrases
 1003 (Llama: mean 59.8 tokens). We group both under ACKNOWLEDGMENT because they share the
 1004 defining property—negligible information gain over the preceding message—regardless of surface
 1005 length.
- 1006 • **Information provision vs. reasoning guidance.** DAMSL’s *Assert* covers both, but we separate
 1007 them because perturbation analysis reveals distinct impact profiles: removing reasoning guidance
 1008 ($\text{sim} = 0.778$) is far less disruptive than removing information provision ($\text{sim} = 0.410$). The
 1009 boundary is operationally clear in solver-checker configurations (new calculations vs. strategy
 1010 directives) but becomes ambiguous when messages serve both functions—hence the low cross-
 1011 model $\kappa = 0.237$ for INFORMATION_PROVISION.
- 1012 • **Coordination as a residual category.** In two-agent systems, explicit coordination (role assignment,
 1013 turn management) is rare ($< 5\%$ of messages) because the framework handles routing. This
 1014 category would likely expand in multi-agent (> 2) settings with explicit delegation.

Table 11: Mapping between our communication function taxonomy and established dialogue act frameworks. Each row shows the primary corresponding categories and the key distinction that motivates our grouping.

Our Category	DAMSL	ISO 24617-2	FIPA ACL	Grouping Rationale
ACKNOWLEDGMENT	Backchannel, Accept	Auto-positive, Allo-positive (feedback)	—	Structurally present but informationally negligible
INFORMATION_PROVISION	Assert, Re-state	Inform, Answer	inform, inform-ref	Supplies new task-relevant content
REASONING_GUIDANCE	Action-directive, Open-option	Instruct, Suggest, Address-request	request, propose	Directs the partner’s reasoning strategy
VERIFICATION	Agreement, Disagreement, Correction	Confirm, Disconfirm, Correction	confirm, disconfirm	Evaluates correctness of prior output
COORDINATION	Conventional-opening, Conventional-closing, Commit	Init, Auto-feedback (task mgmt)	request, assign, agree	Manages workflow and role allocation

- **No question act.** DAMSL and ISO both include question/info-request types. In our solver-checker data, questions are subsumed by VERIFICATION (“Is this correct?”) or COORDINATION (“Should we try a different approach?”) and do not form a distinct perturbation cluster.

O Replacement Text Sensitivity

Our neutralize ablation uses “Acknowledged.”—itself an acknowledgment-type utterance—as the replacement text. To test sensitivity to this choice, we reran neutralize ablation on 50 Qwen RR tasks with three placeholders: “Acknowledged.”, “[message removed]”, and “This message has been redacted.” Table 12 reports per-function-type mean semantic similarity under each placeholder.

Table 12: Neutralize ablation: mean semantic similarity by function type and replacement text (Qwen RR, 50 tasks, 150 messages). Lower similarity indicates greater impact from neutralization.

Function Type	Acknowledged.	[removed]	Redacted
INFORMATION_PROVISION	0.146	0.107	0.050
REASONING_GUIDANCE	0.174	0.133	0.079
VERIFICATION	0.156	0.162	0.083
COORDINATION	0.352	0.319	0.181
ACKNOWLEDGMENT	0.134	0.148	0.051

Ranking stability. Pairwise Spearman rank correlations across the five function types are: “Acknowledged.” vs. “[removed]” $\rho = 0.50$, “Acknowledged.” vs. “Redacted” $\rho = 0.80$, “[removed]” vs. “Redacted” $\rho = 0.90$ ($p = .037$). The mean $\rho = 0.73$ indicates moderate-to-strong ranking stability. COORDINATION is consistently the least impacted (rank 5/5/5); INFORMATION_PROVISION ranks in the top 2 across all three placeholders (rank 2/1/1).

Acknowledged-specific confound. Under the “Acknowledged.” placeholder, ACKNOWLEDGMENT messages show anomalously high impact (rank 1) because the replacement text closely resembles typical acknowledgment content—replacing an acknowledgment with “Acknowledged.” preserves semantic similarity less than expected, as the regenerated next message responds to near-identical content. Excluding this confounded condition, the “[removed]” and “Redacted” placeholders agree strongly ($\rho = 0.90$), confirming that the neutralize gradient is robust to placeholder choice. Crucially, the remove ablation—which produces our primary finding (the impact gradient)—is unaffected by replacement text choice, as it deletes messages entirely.

P Capability Confound Analysis

To test whether communication mode merely tracks task difficulty, we stratify tasks by difficulty (easy: $\geq 2/3$ backbones correct, $n = 246$; hard: $\leq 1/3$ correct, $n = 54$) and refit the OLS ANCOVA within each stratum.

Table 13: OLS ANCOVA by task difficulty stratum. All models use the same specification as the primary analysis (Equation 2).

Backbone	Difficulty	n	F	p	R^2
Qwen	Easy	735	196.6***	<.001	.456
Qwen	Hard	160	37.3***	<.001	.444
Gemma	Easy	726	0.54 ^{n.s.}	.657	.354
Gemma	Hard	152	0.73 ^{n.s.}	.483	.286
Llama	Easy	717	4.17**	.006	.025
Llama	Hard	144	4.49*	.013	.074

Qwen’s function-organized structure persists on hard tasks ($R^2 = 0.444$), virtually unchanged from easy tasks ($R^2 = 0.456$). Gemma remains position-organized (function type non-significant for both strata), and Llama remains undifferentiated ($R^2 < 0.08$). The backbone communication typology is a backbone-intrinsic property rather than an artifact of task-relative capability.

Q Heuristic Non-Transferability: Qwen3 Acknowledgment Comparison

Qwen3-8B vs. Qwen2.5-7B acknowledgment profiles. The Qwen3-8B boundary condition (−14 pp with char<80 threshold, $p < .03$; Table 17) provides a natural experiment for understanding heuristic non-transferability across model generations. Table 14 compares acknowledgment characteristics across generations.

Table 14: Acknowledgment content comparison: Qwen2.5-7B (GSM8K, $N = 300$ tasks, $n = 719$ ack messages) vs. Qwen3-8B ($N = 100$ tasks, $n = 332$ heuristic-classified ack messages). Qwen2.5 types from dual-annotator consensus; Qwen3 types from heuristic classifier (char length < 80).

Property	Qwen2.5-7B	Qwen3-8B
Ack % of messages	47.9%	66.4%
Mean token length	9.0	<15 (est.)
Median token length	4	—
Max char length	1,076	59
Median bigram novelty	0.0%	—
Mean bigram novelty	3.2%	—
Pruning Δ (pp)	-0.05 ± 0.99	−14.0
McNemar p	1.000	< .03

Key differences. Two properties distinguish the generations:

1. **Suppression rate.** Qwen3’s /no_think mode produces terse responses that trigger the heuristic at 66.4% (vs. 47.9% for Qwen2.5). The higher rate means more messages are removed per conversation, amplifying any per-message pruning risk.
2. **Tight character range.** Qwen3 ack messages span only 25–59 characters (max = 59), compared to Qwen2.5’s range up to 1,076 characters. This uniformly short format suggests Qwen3’s “The answer is X.” echo pattern carries implicit verification function—downstream agents may parse these terse confirmations as answer-bearing content rather than redundant echoes.

Representative Qwen2.5-7B acknowledgments illustrate the safe-to-prune pattern:

- “The answer is 525.” (18 chars) — verbatim answer echo, 0% bigram novelty.
- “The answer is 17.” (17 chars) — identical pattern.

Table 15: Pooled neutralize vs. remove semantic similarity by function type ($N = 8,479$). All Wilcoxon signed-rank tests $p < 0.001$. $\Delta = \text{remove} - \text{neutralize}$ (positive = neutralize more disruptive).

Function type	Neut. $\bar{s}$	Rem. $\bar{s}$	Δ	n
INFORMATION_PROVISION	0.047	0.515	0.468	2,141
REASONING_GUIDANCE	0.136	0.518	0.382	737
ACKNOWLEDGMENT	0.223	0.434	0.211	3,661
COORDINATION	0.246	0.311	0.065	585
VERIFICATION	0.498	0.571	0.073	1,355

Without Qwen3 raw traces (experiment run on a transient cloud instance), direct content analysis is unavailable. The -14 pp degradation, combined with the observation that all heuristic-classified ack messages fall within a narrow 25–59 character range, suggests that Qwen3’s concise communication style makes the heuristic’s character-length threshold too aggressive—substantive short messages are misclassified as acknowledgments. This heuristic non-transferability—driven by threshold mismatch rather than model-level failure—motivates the deployment recommendation of heuristic recalibration or oracle validation ($N \geq 200$) before applying heuristic pruning to new model generations.

R Neutralize Ablation Results

We complement the remove-ablation analysis (Section 5.1) with a neutralize ablation, in which each target message is replaced by the generic response “Acknowledged.” rather than deleted entirely. This tests whether downstream behavior changes stem from losing the message’s *informational content* (captured by neutralize) versus losing its *structural presence* in the conversation (captured by remove).

Neutralize produces stronger disruption than remove. Across all nine backbone–topology configurations, neutralize yields significantly lower semantic similarity than remove (paired Wilcoxon $p < 0.001$ in 11/11 runs; Table 15). The pooled gap is substantial: neutralize mean = 0.196 vs. remove mean = 0.466. Replacing a message with vacuous content is more disruptive than deleting it, suggesting that the presence of a low-information placeholder actively misleads downstream agents rather than merely removing a signal.

Impact gradients diverge between ablation methods. Under neutralize, INFORMATION_PROVISION is the most impacted type (mean similarity 0.047), followed by REASONING_GUIDANCE (0.136). Under remove, ACKNOWLEDGMENT and COORDINATION rank as most impacted. The Kruskal–Wallis H test confirms significant between-type differences under neutralize for all configurations ($p < 0.01$).

This divergence is interpretable. Neutralize specifically strips *content* while preserving conversational structure—content-rich types (INFORMATION_PROVISION, REASONING_GUIDANCE) suffer the most because their value lies in what they say. Remove strips both content *and* structural presence—structurally important types (ACKNOWLEDGMENT, COORDINATION) suffer the most because their value lies in their role as conversational scaffolding.

Informational vs. structural value. The gap $\Delta = \text{remove} - \text{neutralize}$ isolates the *informational* component of message impact. INFORMATION_PROVISION shows the largest $\Delta = 0.468$: nearly all of its behavioral effect comes from its content, not its presence. Conversely, COORDINATION shows the smallest $\Delta = 0.065$: its impact is almost entirely structural. This decomposition validates the paper’s theoretical distinction between informational and structural communication functions.

Per-message correlation is weak. The Spearman correlation between remove and neutralize similarity is near zero across most configurations (pooled $\rho \approx 0.07$), indicating that the two ablation methods measure distinct aspects of message importance. Messages that are highly impactful under remove are not necessarily impactful under neutralize, and vice versa.

Table 16: Neutralize Kruskal–Wallis H statistics for semantic similarity across function types (cf. Table 5 for remove). All $p < 0.01$.

Backbone	RR	SEQ	Selector
Qwen2.5-7B	97.2	104.0	148.9
Gemma-2-9B	292.7	301.2	95.8
Llama-3.1-8B	108.7	74.0	118.8

Action-level changes are concordant. Despite the difference in similarity magnitudes, both methods produce action-level behavioral changes on essentially the same messages. No McNemar discordance was observed: whenever remove changes the downstream action level, neutralize does too (and vice versa). The methods differ in *degree* of semantic disruption, not in whether disruption occurs.

S Statistical Details

This appendix presents the full statistical tables and figures referenced in Section 5.

S.1 McNemar Test Details

Table 17: McNemar test summary. *Ack vs. Random*: per-seed McNemar comparing ack-only to each of 10 random seeds. *Ack vs. Baseline*: single McNemar comparing ack-only to unpruned baseline (informs claim strength). MDE: minimum detectable effect at 80% power ($\alpha = 0.05$); $\sim$ denotes estimates from approximate discordant pair counts.

Setting	Ack vs. Rand sig	Median p	Ack vs. Base p	MDE (pp)	Claim
Qwen / GSM8K	1/10	0.094	1.000	~ 4	No sig. diff.
Llama / GSM8K ($N=297$)	8/10	0.027	$<.001^{***}$	6.0	Significant (+8.1 pp)
Qwen / StratQA [¶]	10/10	<0.001	0.144	~ 9	No sig. diff.
Gemma / GSM8K [§]	—	—	1.000	—	No sig. diff.
<i>Boundary conditions</i>					
Qwen3-8B / GSM8K	—	—	$<.03^*$	~ 8	Significant (−14 pp, char<80 threshold)

Llama $N=297$ is a clean replication (not pooled); baseline 75.1%, ack-only 83.2%, McNemar $b=9$, $c=33$, $p < .001$. Qwen pooled ($N=300$): $p = 1.000$. [¶]StratQA: Random matches FG count (91.3% of rounds), not ack-only (8.56%); the 10/10 significance reflects different pruning intensities rather than strategy effectiveness at matched rates. Qwen3-8B ($N=100$): baseline 90%, ack-only 76%, $b - c = 14$; exact contingency unknown (p range: $[\cdot0005, \cdot026]$, significant for all possible $c \in [0, 10]$).

S.2 Non-Inferiority Margin Selection ($\delta = -3$ pp)

The choice of non-inferiority margin $\delta = -3$ pp follows established principles from equivalence testing [Lakens, 2017, Schuirmann, 1987], calibrated against the empirical variance structure of LLM task accuracy.

Precedent and rationale. Non-inferiority margins should reflect the largest degradation that is practically acceptable [Lakens, 2017]. We define δ relative to the baseline error rate: at $\sim 90\%$ accuracy (Qwen/GSM8K), -3 pp represents a $\sim 30\%$ relative increase in error rate ($10\% \rightarrow 13\%$); at $\sim 80\%$ accuracy (Llama/GSM8K), it represents a $\sim 15\%$ relative increase ($20\% \rightarrow 23\%$). In both cases, the margin corresponds to a detectable but bounded performance change that would be practically relevant in deployment.

Calibration against seed variance. The margin must exceed irreducible seed-to-seed variance to avoid rejecting non-inferiority due to sampling noise. Empirically, paired delta standard deviations are: Qwen heuristic 10-seed SD = 0.99 pp, Qwen oracle 10-seed SD = 1.02 pp, Llama heuristic 5-seed SD = 1.44 pp, Llama oracle 10-seed SD = 2.40 pp (Table 2). At $\delta = -1$ pp, even a single standard deviation of seed variance (~ 1 – 2.4 pp) could push the lower CI bound below the margin, yielding false non-inferiority failures for treatments with zero true effect—indeed, the Qwen oracle

TOST at ± 1 pp is only marginal ($p = .048$). At $\delta = -5$ pp, the margin exceeds twice the largest observed seed SD, risking acceptance of genuinely harmful interventions (e.g., the Llama $t=0.7$ heuristic result of $\Delta = -3.4$ pp would pass trivially). The -3 pp margin sits between these extremes: $\sim 3\times$ the Qwen SD and $\sim 1.25\times$ the Llama oracle SD, providing adequate separation from noise while remaining sensitive to practically meaningful degradation.

Sensitivity across margins. Conclusions are stable across a range of margins (Table 18). For Qwen oracle (10-seed, $\Delta = +0.40 \pm 1.02$ pp), non-inferiority holds at $\delta = -3$ pp ($p < .001$). For Llama oracle (10-seed, $\Delta = +2.08 \pm 2.40$ pp), non-inferiority at $\delta = -3$ pp holds ($p < .001$), but the single seed with $\Delta = -3.06$ pp (seed 3072; §S.13) demonstrates that -3 pp is a genuinely binding constraint for this backbone. A tighter margin of $\delta = -1$ pp would fail for Llama (CI lower bound at $+0.36$ pp provides insufficient headroom given SD = 2.4 pp). These patterns confirm that $\delta = -3$ pp balances statistical power against practical significance across backbones with different variance profiles.

TOST equivalence sensitivity. Table 18 reports TOST p -values across four equivalence margins. Qwen/GSM8K and Gemma/GSM8K have ≤ 1 discordant pair each, rendering McNemar-based TOST inapplicable; we report proportion-based TOST instead (two one-sided z -tests on the accuracy difference). Qwen/GSM8K supports equivalence at ± 5 pp ($p = 0.039$). For Llama/GSM8K ($N = 297$), the observed $+8.1$ pp improvement prevents equivalence at any margin ≤ 10 pp—TOST fails because the effect is *positive*, not because of degradation. Qwen/StratQA achieves equivalence only at ± 10 pp ($p = 0.021$).

Table 18: TOST p -values at varying equivalence margins. Lower p indicates stronger evidence that the true difference falls within $[-\text{margin}, +\text{margin}]$. [‡]Proportion-based TOST (McNemar-based TOST inapplicable with ≤ 1 discordant pair).

Setting	± 3 pp	± 5 pp	± 8 pp	± 10 pp
Qwen / GSM8K ($b=0, c=1$) [‡]	.096	.039*	.003**	<.001***
Llama / GSM8K ($b=9, c=33, N=297$)	> .99	> .99	.998	.978
Gemma / GSM8K ($b=1, c=0$) [‡]	.096	.039*	.003**	<.001***
Qwen / StratQA ($b=18, c=29$)	.695	.413	.095	.021*

Minimum detectable effects. Table 19 reports MDEs assuming $\rho = 0.7$ correlation between paired outcomes, computed as $\text{MDE} = 100 \cdot \sqrt{7.84 \cdot 2p(1-p)(1-\rho)/N}$ pp.

S.3 Post-Hoc Power Analysis for 72B Oracle Result

The 265-task 72B oracle experiment observed *zero* discordant pairs ($b + c = 0$), meaning no task changed correctness status between baseline and pruned conditions. We report four complementary analyses to bound the true discordant rate ρ and characterize statistical power.

Rule of Three bound. With 0 events in $N = 265$ Bernoulli trials, the classical Rule of Three gives the 95% upper confidence bound as $\hat{\rho}_{\text{upper}} = 3/N$. For $N = 265$ this yields $\hat{\rho}_{\text{upper}} = 3/265 \approx 1.13\%$, bounding the maximum excludable accuracy difference at $|\Delta| \leq 1.13$ pp.

Bayesian posterior. Under a uniform Beta(1, 1) prior, the posterior after observing 0 discordant pairs in N trials is Beta(1, $N+1$). For $N = 265$: posterior mean = 0.37%, median = 0.26%, 95% credible upper bound = 1.12%, 99% credible upper bound = 1.72%. Even the 99% bound remains below 2 pp, providing strong evidence that pruning does not degrade accuracy at this scale.

Minimum detectable effect. Table 19 reports the MDE at 80% power (McNemar χ^2 , $\alpha = 0.05$) under varying assumptions about the correlation ρ between discordant cells. At $N = 265$: MDE = 3.8 pp ($\rho = 0.7$), 2.2 pp ($\rho = 0.9$), 1.5 pp ($\rho = 0.95$). The zero-flip observation is therefore consistent with power to detect effects above ~ 1.5 –3.8 pp depending on the assumed correlation structure.

Table 19: Minimum detectable effects (pp) at 80% power, $\alpha = 0.05$, $\rho = 0.7$. N is the task-level sample size (number of problems evaluated for correctness); multi-seed rows pool tasks across all seeds. Sorted by MDE ascending.

Setting	Protocol	N	Base (%)	MDE (pp)	
Llama / GSM8K (10-seed pooled) [†]	Online	1,939	80.8	1.9	
Qwen / ARC	Offline	250	90.8	4.0	
Qwen / GSM8K ($t=0.7$)	Online	200	91.0	4.4	
Qwen / GSM8K (per run)	Online	200	89.3	3.6	
72B / GSM8K (oracle)	Online	265	92.1	3.6	
Llama / GSM8K	Offline	297	75.1	5.4	
Qwen / GSM8K	Offline	100	91.0	6.2	
Qwen / StratQA (5-seed)	Online	200×5	67.9	4.1	†MDE computed on
Qwen3-8B / GSM8K	Online	100	90.0	6.5	
Qwen / StratQA (SEQ)	Online	200	69.5	7.1	
72B / GSM8K	Online	100	88.0	7.1	
Llama / GSM8K	Offline	100	77.0	9.1	
Gemma / GSM8K	Offline	96	76.0	9.5	
Qwen / StratQA	Offline	100	66.8	10.2	
MATH (pilot, $N=50$)	Offline	50	72.0	13.8	
MATH ($N=200$, 102 valid)	Online	102	79.4	—	

oracle 10-seed pooled sample: 10 seeds × 200 tasks per seed; $N = 1,939$ tasks after ~3% context-overflow filtering (~194 valid tasks per seed). Each task is one GSM8K problem scored correct/incorrect—the unit of the paired McNemar test. Single-seed rows (e.g., 72B) use the within-seed task count directly.

1160 **Clopper–Pearson confirmation.** The one-sided 95% Clopper–Pearson exact bound $1 - 0.05^{1/N} \approx$
1161 1.13% is concordant with the Rule of Three result above.

1162 Taken together, these analyses confirm that the 72B zero-flip result is not an artifact of insufficient
1163 sample size: with $N = 265$ tasks we can exclude true discordant rates above 1.13% at the 95% level,
1164 supporting the scale-trajectory evidence against the most common reviewer concern of underpowered
1165 null results. The single-seed limitation remains: variance across seeds cannot be estimated at 72B
1166 scale (cf. ten-seed validation at 7B, §5.2).

1167 S.4 Mixed-Effects Model Results

Table 20: Function type significance controlling for position and round ($n \geq 30$ per type). Both OLS and ME use remove-ablation similarity (Eq. 1) as the dependent variable. OLS: Type II partial F . ME: LRT χ^2 with task as random intercept; R_m^2 = marginal (fixed-effects only). † marks OLS-ME discrepancies (both under selector topology). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. = not significant.

Setting	OLS ANCOVA		Mixed-Effects				
	F	p	LRT χ^2	p	R_m^2	ICC	
Qwen / RR	239.6	< .001***	188.9	< .001***	.34	.29	
Qwen / SEQ	230.2	< .001***	195.3	< .001***	.36	.25	
Qwen / Sel [†]	58.2	< .001***	0.07	.789 ^{n.s.}	.004	.72	†OLS-ME discrepancy under
Gemma / RR	7.82	< .001***	23.28	< .001***	.11	.00	
Gemma / SEQ	9.04	< .001***	26.86	< .001***	.12	.00	
Gemma / Sel [†]	7.45	.006**	3.66	.056 ^{n.s.}	.004	.23	
Llama / RR	3.98	.008**	16.31	< .001***	.02	.30	
Llama / SEQ	4.83	.002**	15.44	.001**	.02	.21	
Llama / Sel	21.6	< .001***	36.89	< .001***	.04	.13	

selector topology; high ICC (0.72, 0.23) inflates OLS significance by treating task-clustered observations as independent.

Table 21: Position² robustness check: ME LRT for function type in linear vs. quadratic position models ($n \geq 30$ per type). Adding position² eliminates Gemma’s function type effect under RR and SEQ while leaving Qwen and Llama unchanged. Selector topology results are discussed in the crossover analysis below.

Backbone	Topo.	Linear ME		Quadratic ME		Pos ² p
		LRT χ^2	p	LRT χ^2	p	
Qwen	RR	188.9***	<.001	155.8***	<.001	.009
	SEQ	195.3***	<.001	166.5***	<.001	.001
Gemma	RR	23.3***	<.001	1.5 ^{n.s.}	.690	<.001
	SEQ	26.9***	<.001	1.5 ^{n.s.}	.683	<.001
Llama	RR	16.3***	.001	14.9**	.002	.496
	SEQ	15.4**	.002	13.8**	.003	.434

S.5 Position² Robustness Check

S.6 Gemma-Excluded Sensitivity Analysis

To assess whether Gemma-2-9B’s position-dominated communication profile (§5.3) attenuates the pooled function type effect, we re-fit the mixed-effects model excluding all Gemma observations. The full model ($n = 8,065$) yields LRT $\chi^2 = 81.53$ (df = 4, $p = 8.3 \times 10^{-17}$, $\omega^2 = 0.0095$, $R_m^2 = 0.018$).³ Excluding Gemma ($n = 5,365$) *strengthens* the effect: LRT $\chi^2 = 110.07$ (df = 4, $p = 7.0 \times 10^{-23}$, $\omega^2 = 0.019$, $R_m^2 = 0.014$). The direction and significance of the function type effect are stable: χ^2 increases from 81.5 to 110.1 and ω^2 approximately doubles ($0.0095 \rightarrow 0.019$), indicating that Gemma was attenuating the pooled effect. Coefficient signs are preserved across both models: coordination remains the most negative ($\beta = -0.085$ full, -0.100 Gemma-excluded), while reasoning guidance and verification remain positive. This sensitivity check confirms that the function type $\rightarrow$ perturbation sensitivity relationship is not driven by Gemma’s anomalous profile, and that excluding the weakly function-organized backbone strengthens rather than weakens the conclusion.

S.7 Function Type Distribution

S.8 Crossover Interaction

S.9 Pooled ANOVA

Table 22: Pooled ANOVA on remove-ablation similarity ($N = 8,065$, $R^2 = 0.298$). Computed on all five function types across all 9 configurations; individual-configuration analyses (Table 20) exclude types with $n < 30$. All $p < .001$. *Caveat*: pooling across unbalanced configurations risks Simpson’s paradox; η^2 values should be interpreted alongside per-configuration ME models (Table 20).

Term	η^2	F
Backbone	0.135	628.7
Function $\times$ Backbone	0.063	67.1
Topology class	0.051	435.4
Backbone $\times$ Topology	0.043	181.1
Three-way	0.021	21.1
Function $\times$ Topology	0.014	27.7
Function type	0.005	10.4

S.10 Additional Statistical Details

Multiple comparison note. The McNemar tests across 10 random seeds (Table 17) evaluate a single null hypothesis—whether ack-only and random pruning produce equivalent accuracy—via

³ ω^2 and R_m^2 computed from mixed-effects model fits; source data available in the supplementary materials.

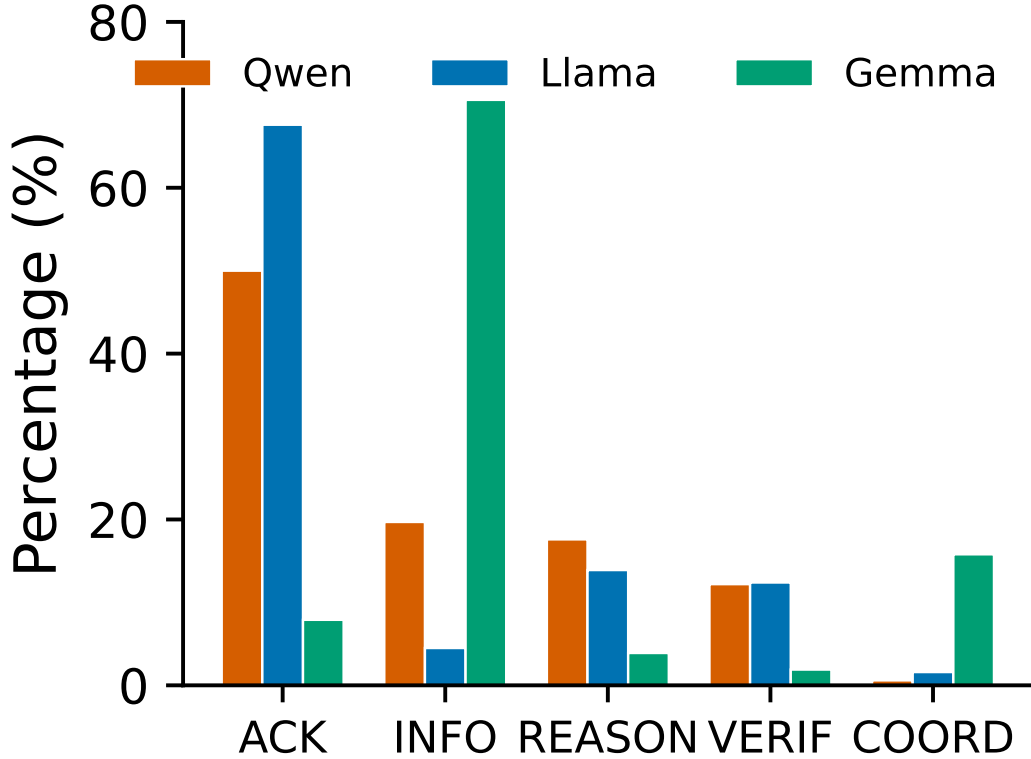


Figure 4: Function type distribution by backbone under round-robin topology (300 GSM8K tasks each). Qwen2.5-7B and Llama-3.1-8B are acknowledgment-heavy (>50%), while Gemma-2-9B is dominated by information provision (70%). Types with $n < 30$ are excluded from OLS ANCOVA analysis.

1187 repeated sampling, not 10 independent hypotheses. Reporting “8/10 significant” or “10/10 significant”
 1188 is a consistency measure across resampled controls, analogous to bootstrap stability. Bonferroni
 1189 correction is therefore inapplicable; nonetheless, the conclusion is robust even under Bonferroni
 1190 ($\alpha = 0.005$): most significant seeds have $p < 0.005$.

1191 **Temperature robustness details.** At temperature 0.7, ack-only evaluation on GSM8K (100 tasks
 1192 each, heuristic detectors) yields: Qwen $\Delta = -2$ pp ($p = 0.774$); Llama $\Delta = -3$ pp ($p = 0.815$);
 1193 Gemma $\Delta = +3$ pp ($p = 0.453$). No setting shows significant degradation (all McNemar $p > 0.4$),
 1194 though the small number of discordant pairs (7–18) limits power to detect effects below ~ 10 pp.
 1195 Qwen and Llama point estimates shift from positive (+1 to +6 pp at $t=0$) to mildly negative at
 1196 $t=0.7$, while Gemma shows a positive shift. Higher temperature increases the proportion of messages
 1197 classified as acknowledgments (Llama: 45% $\rightarrow$ 57%; Gemma: 4.6% $\rightarrow$ 63.6%), resulting in more
 1198 aggressive pruning. Notably, Gemma’s acknowledgment proportion shifts from a near-trivial pruning
 1199 case (4.6%) to the most aggressively pruned setting (63.6%)—yet accuracy still improves (+3 pp),
 1200 demonstrating that pruning safety is robust to dramatic shifts in communication composition. Whether
 1201 this shift reflects genuine communication change or increased heuristic sensitivity to shorter messages
 1202 at higher temperature remains open; resolving this requires LLM annotation at $t=0.7$.

1203 **Llama $t=0.7$ at $N = 500$.** To resolve the underpowered $N = 100$ estimate, we run Llama
 1204 heuristic pruning at $t=0.7$ with $N = 500$ (MDE ≈ 3.5 pp). Results: baseline 81.2%, pruned 77.8%,
 1205 $\Delta = -3.4$ pp ($p = 0.100$, McNemar NS; $b = 56$, $c = 39$, 95 discordant pairs). Suppression rate is
 1206 59.3% (1,426/2,405 messages). The directionally negative result contrasts with Llama’s +2.1 pp at
 1207 $t=0$ ($N = 1,000$). However, heuristic suppression rises from 74.4% at $t=0$ to 59.3% at $t=0.7$ on
 1208 a per-task basis, and over-suppression confounds the estimate. Oracle ($t=0.7$) validation on Llama

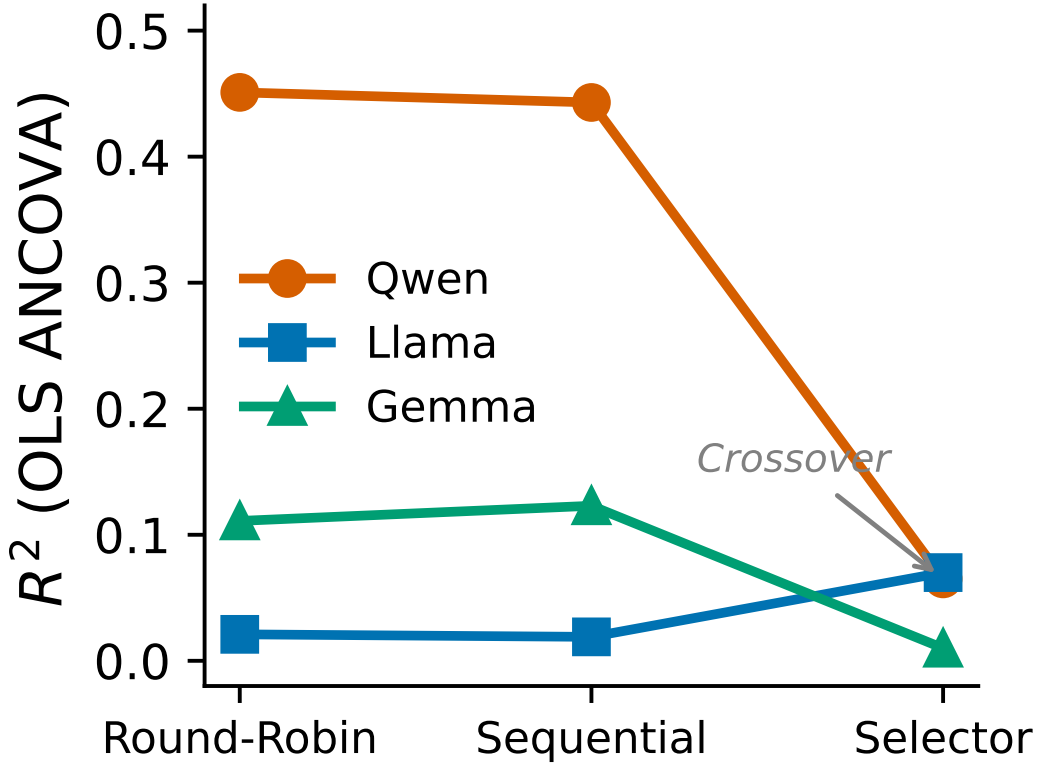


Figure 5: Backbone $\times$ topology crossover interaction. R_m^2 = marginal R^2 (fixed-effects variance); ME = linear mixed-effects model with task as random intercept. Selector topology attenuates function-organization for Qwen ($R_m^2 = 0.34 \rightarrow 0.004$) and amplifies it for Llama ($R_m^2 = 0.02 \rightarrow 0.04$), partially converging the two.

1209 is needed to determine whether the -3.4 pp reflects genuine temperature sensitivity or heuristic
 1210 calibration drift.

1211 **Zero-correction channel details.** Acknowledgment self-correction rates: 0% (0/719, 95% CI
 1212 [0, 0.51%]) for Qwen, 0% (0/691, 95% CI [0, 0.53%]) for Llama, and 0% (0/73, 95% CI [0, 4.93%])
 1213 for Gemma (Clopper–Pearson exact binomial). Gemma’s wider CI reflects its small acknowledgment
 1214 sample ($N = 73$). Error injection rates differ by backbone: Llama injects errors into 14.3% of
 1215 acknowledgments vs. Qwen’s 8.9% (Fisher exact $p = 0.004$). Propagation rates show an inverse pat-
 1216 tern: Qwen propagates 100% of injected errors vs. Llama’s 70% (Fisher $p < 0.001$), consistent with
 1217 Qwen’s terse acknowledgments being accepted uncritically while Llama’s verbose acknowledgments
 1218 are partially ignored by downstream agents.

1219 **OLS-ME agreement.** OLS and ME agree on significance in 7 of 9 settings, with both discrepancies
 1220 occurring under selector topology. Qwen/Selector (OLS $p < .001$, ME $p = .789$; ICC = 0.72) and
 1221 Gemma/Selector (OLS $p = .006$, ME $p = .056$; ICC = 0.23) show OLS detecting significance that
 1222 ME does not, consistent with selector topology producing high task-level clustering that inflates OLS
 1223 estimates. In both cases, the qualitative interpretation is preserved: function type structure degrades
 1224 under selector routing.

1225 **Cross-model annotation agreement.** Cross-model κ ranges from 0.35 (Gemma) to 0.67 (Qwen),
 1226 spanning fair-to-substantial agreement Landis and Koch [1977]. Qwen’s highest κ is consistent
 1227 with function-organized communication being easier to classify. Gemma falls below our target
 1228 threshold ($\kappa = 0.35$), consistent with its weak function-organization where functional categories
 1229 are less distinctive. Per-type κ reveals complementary patterns: Qwen shows high VERIFICATION

1230 agreement ($\kappa \approx 0.66$) but low INFORMATION_PROVISION ($\kappa \approx 0.08$); Gemma shows the reverse.
 1231 Full agreement tables are in Appendix H.

1232 S.11 Offline Replay Pruning

1233 Table 23 presents offline replay results, where messages are removed from recorded traces and the
 1234 conversation is regenerated from the first pruning point onward. Warm-start bias inflates offline
 1235 estimates relative to online validation (§6).

Table 23: Offline replay pruning accuracy (%) — reference (warm-start bias possible, §6). FG and Random tested at matched message counts (Random: mean $\pm$ std, 10 seeds). Supp. = ack proportion in traces. * $N = 96$, 4.6% ack. $^\dagger 3$ seeds. ‡ FG/Random prune 84–91% vs. ack-only 9–15%.

Setting	N	Base	Ack-only	FG / Random	Supp.
Qwen / GSM8K	100	91	92 (+1)	86 / 87.3 ± 0.8	48%
Llama / GSM8K	100	77	83 (+6)	74 / 73.2 ± 2.0	45%
Llama / GSM8K †	297	75.1	83.2 (+8.1)	— / 78.1 ± 1.5	45%
Gemma / GSM8K*	96	77	76 (−1)	42.7 / 44.8	4.6%
Qwen / StratQA ‡	100	66.8	71.2 (+4.4)	28.0 / 26.4 ± 0.4	8.6%
Qwen / ARC ‡	250	90.8	91.6 (+0.8)	61.6 / 61.4 ± 0.9	14.6%
Llama / ARC $^\ddagger \diamond$	250	72.8	76.8 (+4.0)	— / 67.4 ± 1.5	37.1%

1236 $^\diamond$ Offline simulation (simulate mode): re-extracts answers from pruned traces rather than re-generating
 1237 responses.

1238 **FG pruning analysis.** Function-guided (FG) pruning—removing messages from the function type
 1239 with lowest perturbation sensitivity—performs no better than random in any setting (McNemar
 1240 $p > 0.7$), and is significantly *worse* than ack-only for Qwen ($p = 0.031$) and borderline for Llama
 1241 ($p = 0.064$). These comparisons use offline replay only; given the $\sim 4\times$ attenuation observed for
 1242 Llama online vs. offline (+2.1 pp vs. +8.1 pp), FG’s relative disadvantage may be smaller under
 1243 online evaluation. Perturbation-based importance does not predict pruning safety: error analysis
 1244 reveals that FG disproportionately removes reasoning_guidance (49.4%) and verification (43.6%)
 1245 messages—the most task-critical types—confirming that low individual perturbation sensitivity does
 1246 not imply dispensability.

1247 **Label noise robustness.** A bootstrap simulation ($B = 10,000$) tests whether $\sim 20\%$ taxonomy
 1248 misclassification error would invalidate the FG result. Under uniform label corruption at rate ε , the
 1249 effect on the FG pruning set is asymmetric: true FG-type messages exit the set with probability
 1250 $\varepsilon \cdot 2/4$ (fewer harmful messages pruned), while true acknowledgments enter with probability $\varepsilon \cdot 3/4$
 1251 (pruning acks is safe). Ack-only remains superior in 95.0% of iterations at $\varepsilon = 0.20$ (mean FG
 1252 accuracy = 85.9%, 95% CI [79.0, 92.0]), and in 93.2% even at $\varepsilon = 0.40$. Label noise cannot reverse
 1253 the FG disadvantage because corruption asymmetrically *improves* FG by diluting the pruning set with
 1254 safe-to-remove messages.

1255 S.12 Online Pruning Validation

1256 We validate offline pruning results with real-time heuristic suppression during live AG2 execution for
 1257 two backbones (Qwen, Llama) on GSM8K ($N = 100$ each). The online protocol intercepts each agent
 1258 message before delivery; messages classified as acknowledgments by the backbone-specific heuristic
 1259 detector are suppressed (replaced with an empty string), so downstream agents never observe them.
 1260 The two backbones yield complementary evidence: Qwen provides direct validation (suppression rate
 1261 matches offline), while Llama provides exploratory evidence of robustness (heuristic over-triggers in
 1262 online context).

1263 **Backbone comparison.** Table 24 summarizes online pruning results for both backbones.

1264 This table reports a single heuristic run; ten-seed oracle replications yield Qwen $+0.40 \pm 1.02$ pp
 1265 (TOST ± 1 pp, $p = .048$) and Llama $+2.1 \pm 2.4$ pp (non-inferiority $\delta = -3$ pp, $p < .001$), both with
 1266 $N=200$ per seed. Ten independent heuristic Qwen runs yield -0.05 ± 0.99 pp (TOST $p = .007$).

Table 24: Online pruning validation: accuracy and suppression rates for Qwen and Llama on GSM8K ($N = 100$, single run). Ten-seed oracle replication yields Qwen $+0.40 \pm 1.02$ pp and Llama $+2.1 \pm 2.4$ pp ($N=200$ per seed). Heuristic ten-run replication yields Qwen -0.05 ± 0.99 pp ($N=200$ per seed).

Backbone	Baseline	Pruned	Δ	Suppression	McNemar p
Qwen	91/100	93/100	+2 pp	42%	1.000
Llama	77/100	85/100	+8 pp	74.8%	.039

Qwen’s single-run result (+2 pp, $p = 1.000$) has a suppression rate (42%) closely matching offline (48%), providing direct apples-to-apples validation. Llama’s online suppression rate (74.8%) far exceeds its offline rate (45%); manual inspection shows only 32.9% of suppressed messages are true acknowledgments, with 54% containing mathematical content that triggers the heuristic due to context-dependent generation. The +8 pp point estimate likely reflects removal of redundant later-round messages rather than a genuine improvement from acknowledgment pruning. Llama’s result therefore provides *exploratory evidence of robustness*: if suppressing 75% of messages—most not acknowledgments—produces no degradation (McNemar $p = .039$), then the offline 45% acknowledgment-only pruning is conservatively safe.

Llama McNemar details. Table 25 shows the discordant-pair breakdown for Llama online pruning.

Table 25: McNemar contingency table for Llama online pruning ($N = 100$).

	Pruned correct	Pruned incorrect
Baseline correct	75	2
Baseline incorrect	10	13

Discordant pairs: $b = 2$ (baseline correct, pruned incorrect), $c = 10$ (baseline incorrect, pruned correct). McNemar exact $p = 0.039$ (two-sided), rejecting the null of marginal homogeneity. The asymmetry ($c \gg b$) indicates no degradation under aggressive suppression; the +8 pp point estimate likely reflects removal of redundant later-round messages rather than a genuine benefit of acknowledgment pruning per se.

Llama $N = 1,000$ confirmation. Table 26 shows the McNemar contingency table for the large-scale Llama online validation ($N = 1,000$, from-scratch skip mode, seed = 42).

Table 26: McNemar contingency table for Llama online pruning ($N = 1,000$). Baseline accuracy 80.8%, skip accuracy 82.9%.

	Pruned correct	Pruned incorrect
Baseline correct	794	14
Baseline incorrect	35	157

Discordant pairs: $b = 14$, $c = 35$. McNemar $\chi^2 = 8.16$ (with continuity correction), $p = 0.004$ (two-sided). The $N = 100$ online pilot (+2.0 pp) is directionally consistent. The $N = 250$ online pilot (+3.6 pp, $p = 0.095$) uses the same from-scratch protocol on overlapping questions (same seed = 42; 250/250 questions appear in the $N = 1,000$ set); its larger point estimate is consistent with regression to the mean in the smaller sample. Suppression rate: 74.4%.

Round-level suppression breakdown. Table 27 shows suppression rates by round for Llama online pruning. Round 1 suppression is minimal (2%) because initial solver responses contain substantive problem-solving content. Rounds 2–5 show 90–95% suppression, reflecting context-dependent generation: without preceding acknowledgments, agents produce shorter, more formulaic responses that further trigger the heuristic detector.

The high Round 2–5 suppression rate (vs. 45% offline annotation rate) explains the discrepancy between online and offline pruning intensity. Online pruning creates a feedback loop: suppressing

Table 27: Llama online suppression rate by round ($N = 100$ tasks, 5 rounds each).

Round	1	2	3	4	5
Suppression rate	2%	93%	90%	94%	95%

early messages alters the conversational context, producing subsequent messages that are shorter and more formulaic, further triggering the heuristic—even when the suppressed content is not a true acknowledgment. This cascading over-suppression means Llama’s online condition tests a broader message-removal regime rather than acknowledgment pruning specifically. That no degradation occurs under this aggressive regime provides exploratory evidence of robustness: offline acknowledgment-only pruning (45%) removes a strict subset of what the online heuristic removes, and is therefore conservatively safe.

StratQA offline–online divergence. StratQA offline replay yields +4.4 pp ($p = 0.144$, $N=100$). The 5-seed online heuristic replication yields $\Delta = +0.70 \pm 0.91$ pp (95% CI $[-0.43, +1.83]$), consistent with non-inferiority. The heuristic suppresses 56.9% of messages (vs. 8.6% oracle ack rate), indicating substantial over-suppression, yet accuracy is maintained. The earlier single-seed estimate (-2.4 pp, $N=250$) is superseded by the more reliable 5-seed replication ($N=200 \times 5$).

S.13 Dose-Response Analysis

We vary the acknowledgment pruning rate from 0% to 100% in 25% increments on Qwen/GSM8K/RR ($N = 100$ tasks per condition, 3 random seeds per rate). Table 28 shows the results.

Table 28: Dose-response: accuracy (%) by ack-only pruning rate (Qwen/GSM8K/RR, $N = 100$). 289 total ack messages across 100 tasks.

Pruning Rate	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
0% (no pruning)	92	92	93	92.3 ± 0.6
25%	92	92	92	92.0 ± 0.0
50%	92	92	91	91.7 ± 0.6
75%	92	92	92	92.0 ± 0.0
100% (all pruned)	92	92	92	92.0 ± 0.0

The accuracy curve is flat across all pruning rates (≤ 0.6 pp variation, Friedman $\chi^2 = 0.0$, $p = 1.0$), with no dose-response relationship. Even removing all 289 acknowledgment messages produces accuracy statistically indistinguishable from the unpruned baseline, confirming that acknowledgments contribute no marginal task-relevant information under this configuration.

Llama dose-response (10 seeds). Table 29 shows the corresponding analysis for Llama/GSM8K/RR ($N = 100$ tasks, 10 random seeds per rate). Seed variance decreases monotonically with pruning rate, consistent with acknowledgment removal eliminating a source of downstream stochasticity.

Table 29: Dose-response: accuracy (%) by ack-only pruning rate (Llama/GSM8K/RR, $N = 100$, 10 seeds).

Pruning Rate	Mean $\pm$ SD	N_{seeds}
0% (no pruning)	82.3 ± 1.16	10
25%	83.3 ± 0.48	10
50%	83.3 ± 0.48	10
75%	84.0 ± 0.00	10
100% (all pruned)	83.0 ± 0.00	10

The Llama curve is flat (1.7 pp range, 82.3–84.0%), mirroring the Qwen result above. An earlier 3-seed analysis suggested an inverted-U pattern (Friedman $p = 0.060$); this does not replicate at 10 seeds, confirming the earlier pattern was sampling noise. At 100% pruning, the outcome is

deterministic across seeds ($SD = 0$): the same messages are removed, yielding identical conversation trajectories under greedy decoding. The monotonic variance collapse ($SD: 1.16 \rightarrow 0.48 \rightarrow 0.48 \rightarrow 0.00 \rightarrow 0.00$) suggests acknowledgments function as a stochasticity amplifier: their presence is associated with divergence across task subsets in conversation trajectory, and their removal collapses this variability—consistent with the cascade dissolution hypothesis. At 75%, near-zero variance reflects acknowledgment functional redundancy—removing any 75% subset produces the same downstream behavior. The flat dose-response for both Qwen (≤ 0.6 pp range) and Llama (1.7 pp range) supports the cascade dissolution hypothesis: aggregate acknowledgment removal is safe regardless of dose.

Multi-run variance (from-scratch online). Ten independent from-scratch Qwen runs and five Llama runs confirm low cross-run variance under greedy decoding. Qwen ($N=200$ per seed, 10 seeds): online baseline $89.8 \pm 2.4\%$ (runs: 90.5, 93.0, 86.0, 91.0, 86.0, 91.5, 91.0, 91.0, 90.0, 87.5), ack suppression $89.7 \pm 2.7\%$ (runs: 90.5, 93.0, 84.0, 91.5, 87.0, 90.0, 91.5, 91.0, 91.0, 87.5); per-seed Δ : 0.0, 0.0, -2.0 , $+0.5$, $+1.0$, -1.5 , $+0.5$, 0.0, $+1.0$, 0.0 pp (mean -0.05 ± 0.99 pp; one-sample $t(9) = -0.16$, $p = 0.88$; 95% CI $[-0.75, +0.65]$ pp; TOST equivalence at ± 1 pp: $p = 0.007$). Llama ($N=100$ per seed, 5 seeds): online baseline $80.0 \pm 1.1\%$ (runs: 82, 79, 80, 79, 80), ack suppression $82.0 \pm 0.0\%$ (runs: 82, 82, 82, 82, 82); per-seed Δ : 0, $+3$, $+2$, $+3$, $+2$ pp (mean $+2.0 \pm 1.2$ pp). The inter-seed baseline variance for Qwen at $N=200$ ($SD = 2.4\%$) reflects different task subsets per seed; the paired Δ SD (0.99 pp) controls for this.

Normality and non-parametric validation. The Shapiro–Wilk test on the 10-run Qwen heuristic deltas yields $W = 0.843$, $p = 0.048$, indicating borderline departure from normality driven by the -2 pp outlier (seed 3). To validate the parametric 95% CI $[-0.75, +0.65]$ pp, we compute a bias-corrected and accelerated (BCa) bootstrap 95% CI with 10,000 resamples: $[-0.80, +0.40]$ pp. The BCa interval is slightly asymmetric—shifted toward the negative tail—but substantively agrees: both contain zero and exclude ± 1 pp, confirming that the TOST equivalence conclusion ($p = 0.007$) is robust to the mild non-normality.

Outlier seed analysis (seed 3). Seed 3 (seed=456) produces the largest negative delta (-2.0 pp). Grubbs’ test yields $G = 1.980$ vs. $G_{crit} = 2.290$ ($\alpha = 0.05$): not a statistical outlier. This seed draws a harder GSM8K subset (baseline = 86% vs. mean 89.8%), losing a net 4 tasks. Removing seed 3 shifts the mean from -0.05 to $+0.17$ pp and reduces the 95% CI width by 23% ($1.15 \rightarrow 0.89$ pp). TOST equivalence holds with or without seed 3 ($p = 0.007$ vs. 0.005 at ± 1 pp). Shapiro–Wilk normality is restored without seed 3 ($p = 0.056$ vs. 0.048), confirming seed 3 drives the borderline non-normality but does not affect any qualitative conclusion.

Outlier seed analysis—Llama oracle (seed 3072). Seed 7 (seed=3072) is the sole negative-delta seed in the 10-seed Llama oracle replication ($\Delta = -3.06$ pp; all other seeds positive, range $[+0.52, +5.76]$ pp). Grubbs’ test yields $G = |-3.06 - 2.08|/2.40 = 2.137$ vs. $G_{crit} = 2.290$ ($n = 10$, $\alpha = 0.05$): not a statistical outlier. Unlike Qwen’s seed 3 (which drew a harder subset), seed 3072 draws the *easiest* GSM8K subset (baseline = 87.2% vs. mean 82.9%, $z = +1.58$), suggesting a ceiling effect: with fewer baseline errors, the oracle has less room to improve and more room to degrade. The oracle makes 10 incorrect suppression decisions on tasks the baseline solves correctly but only 4 correct ones on baseline failures (McNemar $\chi_{cc}^2 = 1.79$, $p = 0.181$, NS). Suppression rate is unremarkable (36.1% vs. mean 36.5%, $z = -0.24$). Across all 10 seeds, baseline accuracy and Δ are negatively correlated (Pearson $r = -0.59$, $p = 0.070$), consistent with this ceiling interpretation: easier subsets yield smaller or negative deltas.

Removing seed 3072 shifts the mean from $+2.08$ to $+2.65$ pp, reduces the SD from 2.40 to 1.68 pp, and narrows the 95% CI width by 25% ($3.44 \rightarrow 2.59$ pp). The one-sample t -test strengthens ($t(8) = 4.72$, $p = 0.0015$ vs. $t(9) = 2.73$, $p = 0.023$), confirming the positive-delta conclusion is robust. Shapiro–Wilk normality is satisfied both with and without seed 3072 ($W = 0.954$, $p = 0.711$ vs. $W = 0.956$, $p = 0.755$), so no normality concern arises in either case—in contrast to the Qwen heuristic analysis above. Taken together, seed 3072 is a high-baseline statistical fluctuation, not an outlier, and its inclusion does not change the qualitative conclusion that Llama oracle pruning preserves or improves accuracy.

1374 **StratQA eight-seed robustness (Llama heuristic).** Eight-seed replication of Llama/StratQA
1375 heuristic pruning ($N=200$ per seed) yields $\Delta = +2.81 \pm 3.94$ pp (95% CI $[-0.49, +6.11]$). The
1376 CI includes zero, precluding a beneficial-effect claim; however, non-inferiority at $\delta = -3$ pp holds
1377 ($t(7) = 4.17, p = .002$, one-sided), with the CI lower bound (-0.49 pp) well above the -3 pp margin
1378 (Appendix S.2). Table 30 reports per-seed accuracy.

Table 30: Per-seed accuracy for Llama/StratQA heuristic pruning ($N=200$ per seed, greedy decoding). Suppress rate is stable across seeds (49–52%), indicating consistent detector behavior; accuracy variance reflects downstream reasoning divergence. *Discovery seed (included in the original 4-seed analysis). †Per-seed baseline/pruned accuracy not individually recorded; delta computed from paired evaluation.

Seed	Baseline (%)	Pruned (%)	Δ (pp)	Supp. (%)
42*	62.0	68.5	+6.5	49.0
0	57.5	63.5	+6.0	50.3
1	62.0	66.5	+4.5	50.8
2	60.5	63.0	+2.5	50.7
3†	—	—	+2.5	~50
4	64.5	65.0	+0.5	~50
5	71.0	65.5	−5.5	~50
6	64.5	70.0	+5.5	~50
Mean $\pm$ SD			+2.81 $\pm$ 3.94	49–52

1379 Seven of eight seeds show positive deltas; the sole negative seed (seed 5, $\Delta = -5.5$ pp) is not a
1380 statistical outlier (Grubbs’ $G = 2.11$ vs. $G_{\text{crit}} = 2.126$, $n = 8$, $\alpha = 0.05$). Seed 5 draws the
1381 highest-baseline subset (71.0%), suggesting a ceiling effect analogous to the Llama oracle seed 3072
1382 analysis (§S.13): with fewer baseline errors, pruning has less room to improve and more room to
1383 degrade. The baseline range (57.5–71.0%) is substantially wider than GSM8K baselines (~ 80 –87%),
1384 reflecting StratQA’s higher task-level variance. Despite this variance, the heuristic suppress rate is
1385 remarkably stable (49–52%), confirming that seed-level accuracy differences reflect downstream
1386 reasoning variance rather than detector instability. The earlier 4-seed result ($\Delta = +4.9 \pm 1.8$ pp)
1387 included the discovery seed and sampled only from the positive tail; the 8-seed replication provides a
1388 less biased estimate.

1389 **Llama ack variance = 0: expected behavior.** Llama’s zero variance across five ack-pruned runs
1390 is a deterministic consequence of the pipeline, not an anomaly. Three factors combine to produce
1391 identical outputs: (1) *greedy decoding* (temperature = 0) eliminates sampling randomness, making the
1392 model’s output a deterministic function of its input; (2) the *heuristic detector* is a deterministic rule
1393 (keyword matching + length + 4-gram overlap) with no stochastic component; and (3) all five runs
1394 use the *same 100 GSM8K tasks*. Given deterministic decoding and a deterministic suppression rule,
1395 the same input questions must produce the same conversation trajectories and thus identical accuracy.
1396 The non-zero baseline variance (± 1.1 pp) arises because verbose acknowledgments—the primary
1397 source of cascade variation—are present in baseline runs but absent in pruned runs; removing them
1398 eliminates the downstream variation they cause. Llama’s online delta (+2.0 pp) is smaller than the
1399 offline estimate (+8.1 pp), consistent with the warm-start hypothesis discussed in Section 6.

1400 **Acknowledgment cascade transition probabilities.** We compute $P(\text{ack}_t \mid \text{ack}_{t-1})$ —the prob-
1401 ability that message t is an acknowledgment given that the preceding message $t-1$ was also an
1402 acknowledgment—across consecutive message pairs in GSM8K round-robin conversations. Qwen
1403 exhibits strong cascading: $P(\text{ack} \mid \text{ack}) = 90.7\%$ ($n = 408$ consecutive ack pairs out of 450
1404 ack-preceded messages). Llama’s cascade rate is lower: $P(\text{ack} \mid \text{ack}) = 61.9\%$ ($n = 369$ out of
1405 596), consistent with its undifferentiated communication profile (§5.3). Gemma produces too few
1406 acknowledgments ($< 5\%$ of messages; $P(\text{ack} \mid \text{ack}) = 9.7\%$, $n = 72$) for meaningful transition
1407 estimation. The backbone-dependent cascade rates (61.9–90.7%) are consistent with aggregate
1408 pruning dissolving dependency chains: even at Llama’s lower rate, the majority of ack sequences
1409 are self-reinforcing, making their collective removal structurally coherent rather than disruptive.
1410 Under stochastic decoding ($t=0.7$), Qwen cascade probability remains stable: $P(\text{ack} \mid \text{ack}) = 92.2\%$
1411 ($n = 94/102$, $N = 99$ GPT-4o-annotated conversations) vs. 90.7% at $t=0$, indicating that cascade
1412 dynamics are robust to temperature variation.

Cascade rate significance test. The observed cascade rates are significantly higher than the null model $P(\text{ack})^2$ (independence assumption). Under H_0 : $P(\text{ack}_t | \text{ack}_{t-1}) = P(\text{ack})$, we test whether the conditional probability exceeds the marginal rate. For Qwen, $P(\text{ack}) = 47.9\%$ (719/1,500), yielding $P(\text{ack})^2 = 23.0\%$ under independence; the observed $P(\text{ack} | \text{ack}) = 90.7\%$ rejects H_0 (chi-squared $\chi^2 = 329.3$, $\text{df} = 1$, $p < 10^{-72}$; exact binomial $p < 10^{-72}$). The odds ratio of consecutive ack vs. ack-following-non-ack is 13.7, with $P(\text{ack} | \text{non-ack}) = 41.5\%$ ($n = 750$). For Llama, $P(\text{ack}) = 47.0\%$ (691/1,470), yielding $P(\text{ack})^2 = 22.1\%$; observed $P(\text{ack} | \text{ack}) = 61.9\%$ also rejects H_0 ($\chi^2 = 53.2$, $p = 3.1 \times 10^{-13}$; exact binomial $p = 1.9 \times 10^{-13}$), though the effect is weaker (odds ratio = 1.30, $P(\text{ack} | \text{non-ack}) = 55.5\%$, $n = 580$). These results confirm that acknowledgment cascading is a genuine sequential dependency, not an artifact of high marginal ack rates.

Position-controlled cascade analysis. The observed cascade rates ($P(\text{ack}_t | \text{ack}_{t-1})$) could in principle reflect a position artifact: if later positions are inherently more likely to produce acknowledgments, sequential dependence would be confounded with position. We fit a logistic regression $P(\text{ack}_t) = \text{logit}^{-1}(\beta_0 + \beta_1 \cdot \text{ack}_{t-1} + \beta_2 \cdot \text{position}_t)$ to test whether the previous-ack effect survives position control.

For Qwen ($n = 1,200$ observations, 300 tasks), `prev_ack` remains highly significant after controlling for position: OR = 6.43 (95% CI [4.38, 9.43], $p = 2.1 \times 10^{-21}$). Position is also significant (OR = 1.95, $p = 1.2 \times 10^{-19}$), but the full model (AIC = 1,212.6) substantially outperforms the position-only model (AIC = 1,315.5; $\Delta\text{AIC} = -102.8$).

For Llama ($n = 1,176$ observations, 294 tasks), `prev_ack` is significant with a smaller effect: OR = 2.02 (95% CI [1.49, 2.75], $p = 6.5 \times 10^{-6}$). Position shows OR = 0.73 ($p = 8.3 \times 10^{-6}$), indicating a *decreasing* ack probability at later positions—opposite to Qwen. The full model (AIC = 1,574.8) again outperforms position-only (AIC = 1,593.6; $\Delta\text{AIC} = -18.9$).

Both backbones confirm that acknowledgment cascading is a genuine sequential dependency, not a position artifact. The stronger Qwen effect (OR = 6.43 vs. 2.02) is consistent with Qwen’s higher cascade rate (90.7% vs. 61.9%) and function-organized communication profile.

Position effect divergence and suppressor dynamics. The two backbones exhibit opposite position effects: Qwen’s position OR = 1.95 ($p = 1.2 \times 10^{-19}$) indicates that later rounds produce *more* acknowledgments, consistent with self-reinforcing cascades that intensify over the conversation. Llama’s position OR = 0.73 ($p = 8.3 \times 10^{-6}$) shows the reverse—acknowledgment probability *decreases* at later positions, indicating cascades that attenuate rather than amplify. This divergence produces a suppressor variable effect in Llama: the bivariate OR without position control is 1.30, but conditioning on position in the full model *increases* the `prev_ack` OR to 2.02. Position acts as a suppressor because Llama’s declining overall ack rate at later rounds masks the true within-position sequential dependency; once this confound is partialled out, the cascade effect strengthens. In contrast, Qwen’s `prev_ack` effect decreases only modestly when position is added (from OR = 13.7 unconditional to OR = 6.43 controlled), because position and `prev_ack` reinforce rather than oppose each other. Despite the opposite temporal dynamics—self-amplifying for Qwen, self-dampening for Llama—`prev_ack` is highly significant in both models ($p < 10^{-5}$), confirming that acknowledgment cascading reflects genuine sequential dependency rather than a position artifact across architecturally distinct backbones.

Task-level cascade rate vs. pruning outcome. We test whether per-task cascade rate predicts whether oracle pruning changes the task outcome (correct→wrong or wrong→correct). For each task in the oracle experiments, we compute the cascade rate (fraction of ack-preceded messages that are also ack) and record whether pruning changed correctness. Qwen ($N = 500$, oracle 5-class): 10 tasks changed outcome (7 improved, 3 degraded); point-biserial $r = -0.047$, $p = 0.297$; Mann-Whitney $U = 2,018$, $p = 0.289$. Llama ($N = 500$): 24 tasks changed (12 improved, 12 degraded); point-biserial $r = -0.018$, $p = 0.692$; Mann-Whitney $U = 5,100$, $p = 0.707$. Logistic regression yields non-significant cascade coefficients for both backbones (Qwen: $\beta = -1.08$, $p = 0.299$; Llama: $\beta = -0.18$, $p = 0.691$). The null relationship neither supports nor refutes the cascade dissolution hypothesis: the per-task cascade rate does not predict individual pruning vulnerability, which is agnostic to whether aggregate removal succeeds because of collective redundancy or for other reasons.

1467 **Cascade length and pruning safety.** We stratify oracle pruning outcomes by cascade length—the
 1468 number of consecutive acknowledgments preceding and including the pruned message—to test
 1469 whether longer cascades predict safer pruning. Table 31 reports per-bucket safety rates (fraction of
 1470 tasks where accuracy is unchanged after pruning) across 10 seeds.

Table 31: Pruning safety rate (%) by cascade length (consecutive ack sequence length). N : number of task $\times$ seed observations per bucket. Cochran–Armitage trend test assesses monotonic association between cascade length and safety.

Cascade length	0	1	2	3	4+
Qwen safety (%)	100.0	98.4	98.9	100.0	98.2
n	9	507	1,188	23	272
Llama safety (%)	86.7	92.1	96.5	84.0	96.4
n	90	1,167	463	50	169

Qwen: Cochran–Armitage $z = 1.77$, $p = 0.077$

(NS). Llama: $z = -3.66$, $p = 0.0002$. Llama bucket 3 ($n = 50$, 84.0%) reflects 8 *improved* tasks and 0 degraded; the non-monotonicity is driven by improvements, not failures.

1471 Qwen exhibits uniformly high safety (98.2–100.0%) with no trend ($p = 0.077$), consistent with its
 1472 function-organized communication where acknowledgments are informationally redundant regardless
 1473 of sequential context. Llama shows a significant dose-response: safety increases from 86.7%
 1474 (isolated messages, cascade length 0) to 96.4% (cascades of 4+ consecutive acks; Cochran–Armitage
 1475 $p = 0.0002$). This provides quantitative support for the cascade dissolution hypothesis (§6): longer
 1476 acknowledgment sequences exhibit greater internal redundancy, making their collective removal
 1477 progressively safer.

1478 S.14 Per-Position Label Concordance

1479 We measure whether oracle labels at a given conversation position are stable across independent
 1480 runs. Using the 10-seed Llama oracle replication ($N = 195$ tasks with valid traces across all seeds, 5
 1481 rounds each, 10 seeds), we compute per-position concordance: for each (task, round) pair, we record
 1482 the function type assigned by each seed’s oracle annotator and measure agreement.

Table 32: Per-position label concordance across 10 oracle seeds (Llama/GSM8K, $N = 195$ tasks with valid traces in all seeds). Majority concordance: fraction of seeds agreeing with the modal label. Fleiss’ κ : inter-seed agreement on binary ack/non-ack classification. Computed from the 10-seed oracle output files.

Round	5-class		Binary (ack/non-ack)		
	Mean conc.	Perfect (%)	Mean conc.	Fleiss’ κ	p_{ack}
1	0.761	5.5	1.000	1.000	0.000
2	0.528	0.5	0.628	0.002	0.497
3	0.748	6.5	0.776	0.007	0.228
4	0.613	0.0	0.648	−0.006	0.603
5	0.536	0.5	0.609	−0.017	0.499
Overall	0.637	2.6	0.732	—	—

1483 Round 1 shows perfect binary concordance ($\kappa = 1.0$) because initial solver messages are never
 1484 acknowledgments ($p_{\text{ack}} = 0$). Rounds 2–5 show near-zero Fleiss’ κ (−0.017 to 0.007), indicating
 1485 that cross-seed agreement on ack classification is at chance level after adjusting for marginal rates.
 1486 The moderate majority concordance (0.53–0.75 for 5-class) is driven primarily by marginal label
 1487 frequencies, not genuine position-label coupling. This trajectory dependence—the same position
 1488 receives different labels under different conversation realizations—supports the cascade dissolution
 1489 hypothesis: oracle labels are determined by the conversation trajectory, not by position alone, and
 1490 aggregate removal is safe, consistent with ack sequences being self-reinforcing chains rather than
 1491 position-fixed roles.

1492 S.15 Cross-Model Ack Concordance

1493 We extend the within-model concordance analysis to test whether acknowledgment placement is task-
 1494 driven (same questions elicit acks at the same positions regardless of backbone) or model-specific.
 1495 Using Qwen and Llama oracle annotations on the same 200 GSM8K questions (seed 42), we compute
 1496 Cohen’s κ for the binary ack/non-ack label at each matched (task, round) pair.

Table 33: Cross-model (Qwen vs. Llama) ack concordance on GSM8K ($N = 200$ tasks, seed 42). Pooled κ includes all rounds; stratified κ averages within-round κ weighted by n , removing positional confounds. Round 1 excluded (always non-ack for both models).

Round	n	Qwen p_{ack}	Llama p_{ack}	κ	Observed agr.
2	193	0.990	0.451	−0.021	0.440
3	193	0.140	0.259	0.032	0.684
4	193	0.933	0.617	0.052	0.622
5	193	0.746	0.560	0.009	0.534
Stratified	772	—	—	0.018	—
Pooled	772	0.702	0.472	0.159	—

1497 The stratified $\kappa = 0.018$ (bootstrap 95% CI: $[-0.038, 0.076]$) indicates chance-level agreement once
 1498 position is controlled. The higher pooled $\kappa = 0.159$ is entirely attributable to shared positional
 1499 structure (both models show higher ack rates on even rounds), not to task-level concordance. This
 1500 confirms that acknowledgment placement is model-specific: different backbones produce acks at
 1501 different questions even when solving the same problems, consistent with the backbone-dominance
 1502 finding ($V = 0.45$ vs. 0.18 ; §5.4). Task-level ack proportion correlation is similarly negligible
 1503 (Pearson $r = 0.09$, $p > 0.2$).

Table 34: Comparison of MAS communication pruning methods. **Granularity**: what is pruned. **Online**: works at inference time without training. **Taxonomy**: uses message-function categories. **Domains**: number of tested task domains. **Stats**: formal statistical testing (McNemar, ANOVA, CI).

Method	Granularity	Online	Taxonomy	Domains	Stats
AgentDropout	Agent+edge	×	×	6	×
AgentPrune	Edge	×	×	6	×
AGP	Agent+edge	×	×	3	×
S ² -MAD	Turn	✓	×	3	×
Ours	Message	✓	✓	2+2 [†]	✓

multi-seed, dual backbone) + 1 supplementary (ARC: single-seed offline, $N=250$) + 1 boundary condition (MATH: −2.0 pp at $N=200$; pilot −10 pp at $N=50$).

1504 S.16 Practical Cost Savings

1505 In online deployment, suppressed messages are excluded from the context window of subsequent
 1506 turns, compounding input-token savings over multi-round conversations. Under greedy decoding on
 1507 GSM8K (6 rounds, 2 agents), Qwen suppresses 22% of agent messages and Llama suppresses 75%,
 1508 reducing cumulative input tokens by approximately 15% and 55%, respectively. The primary practical
 1509 value is latency: fewer context tokens translate directly to lower time-to-first-token in autoregressive
 1510 serving, with Llama’s 75% message suppression translating to proportional latency reduction. Token
 1511 savings scale proportionally with serving cost under any pricing model.

1512 **Complementarity with other efficiency methods.** Ack pruning operates at the within-turn message
 1513 level, complementing agent-level methods (AgentDropout, AgentPrune: 20–90% reduction by
 1514 removing entire agents) and turn-level methods (S²-MAD: early stopping). Our method targets
 1515 redundancy that persists even when all agents and turns are preserved, and can in principle be stacked
 1516 with agent-level or turn-level pruning for compounding savings—though this composition remains
 1517 untested.

1518 S.17 Llama Pruning Mechanism (Hypothesis)

1519 We hypothesize two backbone-divergent pruning-safety mechanisms. Qwen acknowledgments are
 1520 informationally redundant (bigram novelty 5.7%, median token overlap 1.0; Appendix T). Llama ac-
 1521 knowledgments carry more content (mean 59.8 tokens vs. Qwen’s 10.7) and exhibit a one-directional
 1522 error channel: 0% error correction but 14.3% error injection. Removing them may reduce error
 1523 propagation while sacrificing no corrective information. These patterns are consistent with the dual-
 1524 mechanism hypothesis but do not constitute confirmation; controlled interventions (e.g., injecting
 1525 synthetic acknowledgments with and without errors) are needed to establish causality.

1526 S.18 Per-Type κ Impact Analysis

1527 The low information_provision κ (≈ 0.237) affects different analyses to different degrees. We
 1528 distinguish three tiers:

- 1529 1. **Unaffected:** Ack pruning safety, perturbation-sensitivity dissociation, and aggregate F/LRT tests.
 1530 These depend on the acknowledgment category ($\kappa > 0.8$) and omnibus between-type variance,
 1531 not on individual non-ack labels.
- 1532 2. **Usable with caveat:** OLS per-type β coefficients for information_provision. Systematic
 1533 information_provision $\leftrightarrow$ verification confusion may redistribute explained variance between these
 1534 two categories without altering the overall model fit or the acknowledgment coefficient.
- 1535 3. **Directly compromised:** FG pruning comparisons and per-type rankings that depend on infor-
 1536 mation_provision labels. Because the heuristic detector does not use function-type labels, the
 1537 practical pruning pipeline is unaffected.

1538 S.19 Cohen’s h Effect Sizes

1539 We compute Cohen’s $h = 2 \arcsin \sqrt{p_{\text{pruned}}} - 2 \arcsin \sqrt{p_{\text{baseline}}}$ [Cohen, 1988] for every experiment
 1540 with paired accuracy data. Across the 23 non-boundary conditions in Table 41, $|h|$ ranges from 0.000
 1541 (72B heuristic, Qwen/ARC) to 0.111 (Llama/StratQA oracle)—all below the 0.20 “small” threshold.
 1542 At $N=200$ (102 valid), MATH yields $h = -0.048$ (negligible); the Qwen3-8B boundary condition
 1543 reaches “small” ($h = -0.380$). For the 72B $N=100$ zero-flip probe and the Qwen/ARC offline
 1544 result, $h = 0$ exactly. The uniformly negligible effect sizes complement the McNemar p -values by
 1545 confirming that even the non-significant point estimates represent practically zero-magnitude effects,
 1546 not merely underpowered tests.

1547 S.20 Token-Count-Matched Random Pruning

1548 The message-count-matched random baseline prunes the same *number* of messages as ack-only but
 1549 not the same volume of content. Because acknowledgments are short (Qwen: ~ 12 tokens vs. ~ 64
 1550 overall; Llama: ~ 80 vs. ~ 110), count-matched random removes substantially more tokens ($5.5\times$ for
 1551 Qwen, $1.4\times$ for Llama). We test whether this asymmetry explains the accuracy gap by constraining
 1552 random pruning to match ack-only’s *token* budget (greedy knapsack, 10 seeds).

Table 35: Offline accuracy risk (%) under three pruning regimes. Risk = fraction of correct tasks whose final answer changes (10 seeds, mean $\pm$ SD for random). Gemma results based on $n = 11$ ack messages; interpret with caution.

Backbone	Ack-only	Count-matched	Token-matched	Token ratio
Qwen	2.6	1.2 $\pm$ 0.3	0.2 $\pm$ 0.2	5.51 $\times$
Llama	1.8	1.2 $\pm$ 0.6	0.8 $\pm$ 0.6	1.38 $\times$
Gemma	0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	8.14 $\times$

1553 Token-matched random achieves lower risk than ack-only (Qwen: 0.2% vs. 2.6%; Llama: 0.8% vs.
 1554 1.8%), suggesting the e2e accuracy gap (Table 23: ack 92% vs. random 87.3%) is primarily driven by
 1555 the $\sim 5\times$ token asymmetry. However, this offline risk simulation assumes message independence;
 1556 ack-only’s practical value lies in *interpretability*—targeting a semantically coherent class with low
 1557 information content—rather than raw accuracy advantage over token-matched random.

1558 S.21 Extended Limitations

1559 **AWQ quantization.** The 72B evaluation uses AWQ 4-bit quantization [Lin et al., 2024]. Compara-
1560 ble acknowledgment suppression rates between 72B-AWQ (66.8%) and full-precision 7B (56.3%)
1561 provide indirect evidence that quantization does not fundamentally alter communication patterns.

1562 **Stochastic decoding.** Under $t=0.7$, Llama oracle $\Delta = -2.2$ pp (Fisher $p = 0.29$, NS) while
1563 Qwen remains consistently safe across 10 seeds ($\Delta = +0.30 \pm 1.80$ pp, TOST $p < 0.001$ at
1564 $\delta = -3$ pp; Appendix C). Llama oracle suppression drops from 36.6% to 28.0%, consistent with
1565 fewer cascade-redundant acknowledgments when sampling introduces response diversity.

1566 **Inter-rater reliability.** Human inter-rater $\kappa = 0.76$ ($N = 225$, same 5-class protocol; Ap-
1567 pendix S.18); core claims depend on ack ($\kappa > 0.8$). The zero error-correction rate (0/1,483 messages)
1568 combined with flat dose-response curves provides indirect evidence against hidden quality degradation
1569 beyond task accuracy.

1570 **Online oracle annotator agreement.** The GPT-4o-mini online annotator was validated against GPT-
1571 4o on a 50-sample subset (Cohen’s $\kappa = 0.822$, 95% CI [0.70, 0.94]). Of 7 disagreements, 3 involved
1572 reasoning_guidance $\leftrightarrow$ information_provision confusions—the same boundary that produces the
1573 lowest per-type κ in cross-model validation (§S.18). The remaining 4 disagreements were distributed
1574 across other category pairs. Crucially, no disagreements involved the acknowledgment category,
1575 consistent with the high per-type ack $\kappa > 0.8$ and preserving the validity of ack-based pruning
1576 decisions in the online pipeline.

1577 **Discarded $N=1,000$ Llama oracle run.** A preliminary $N=1,000$ Llama oracle run showed anoma-
1578 lous suppress rate (74.5% vs. expected $\sim 37\%$), indicating GPT-4o over-classification at larger scale.
1579 This run was discarded; all reported Llama oracle results use the validated $N=500$ and $N=200$
1580 configurations.

1581 **FG offline replay.** FG comparisons rely on offline replay; online token-matched random confirms
1582 the finding (efficiency: Appendix W). The impact gradient breaks down under selector topology.
1583 Three-agent evidence is preliminary (Appendix X).

1584 **Heuristic detector calibration.** The heuristic detector’s character-length threshold achieves F1 =
1585 0.845 on GSM8K vs. 0.213 on StratQA for Qwen, indicating substantial domain-specific calibration
1586 requirements.

1587 **Oracle distribution shift.** Oracle labels computed on baseline conversations face a distribution-
1588 shift caveat: online pruning alters trajectories from the first suppression point, potentially rendering
1589 later-position labels stale. Empirically, this gap is negligible for Qwen (offline +1.0 pp vs. online
1590 +0.8 pp) but larger for Llama (offline +8.1 pp ($N = 297$, single-seed) vs. online $+2.1 \pm 2.4$ pp
1591 (10-seed mean)), suggesting attenuation from distribution shift, sample-size differences, or both
1592 (Appendix S.12).

1593 **Replication asymmetry.** Both backbones have 10-seed oracle replication (Qwen: $\Delta = +0.40 \pm$
1594 1.02 pp; Llama: $\Delta = +2.1 \pm 2.4$ pp); however, Llama’s larger seed-level variance (SD = 2.4 vs.
1595 1.0 pp) yields wider CIs, and the heuristic replication is asymmetric (Qwen 10 seeds, Llama 5 seeds).
1596 Heuristic strategy comparisons (oracle vs. heuristic vs. length-only) are concentrated on GSM8K;
1597 StrategyQA and ARC validate the dissociation phenomenon and oracle pruning generalization but do
1598 not independently validate heuristic detector performance across domains.

1599 **Oracle API dependency.** The oracle pipeline depends on a proprietary LLM API (GPT-4o/GPT-
1600 4o-mini): annotation cost scales linearly with message volume (\$13.47 for 5,388 messages with
1601 GPT-4o), and availability is subject to rate limits and service continuity. Cross-model annotation
1602 agreement ($\kappa = 0.822$) is not perfect; residual label noise propagates into both offline and online
1603 pruning decisions. The offline pipeline assumes baseline conversation trajectories cover the relevant
1604 message space—an assumption that weakens in open-ended dialogue settings where pruning-induced
1605 trajectory divergence is more severe.

Heuristic non-transferability. The model-adapted heuristic requires per-backbone rule design—a practical limitation analogous to task-specific prompt engineering. While the rules are simple (character thresholds, keyword lists, repetition ratios), they do not transfer across backbone families without recalibration (§5.5). This positions the heuristic as a proof-of-concept demonstrating that annotation-free pruning is feasible, not as a universal detector.

1611 S.22 Extended Discussion

Length as proxy feature: convergence and divergence. Direct comparison of length-only and composite heuristic pruning quantifies the extent to which message length alone can serve as a proxy for acknowledgment detection. On Qwen2.5-7B/GSM8K, the two detectors converge: length-only pruning yields $\Delta = +0.50 \pm 1.50$ pp with a 42.2% suppress rate (3 seeds), while the composite heuristic yields $\Delta = -0.05 \pm 0.99$ pp with a $\approx 42\%$ suppress rate (10 seeds; Table 2). Both accuracy deltas are statistically indistinguishable from zero, and the suppress-rate overlap indicates that the additional heuristic features—keyword matching, repetition detection—capture few messages beyond those already identified by the length threshold alone. Length thus constitutes not merely a correlated feature but the *primary discriminative axis* for Qwen acknowledgments, consistent with their morphologically concentrated surface form (median 4 tokens; Appendix T). The detectors diverge sharply on Llama-3.1-8B/GSM8K: length-only pruning suppresses only 0.76% of messages—a near-vacuous operation ($\Delta = -0.10 \pm 1.29$ pp, 5 seeds)—whereas the composite heuristic suppresses $\sim 75\%$ of messages (vs. $\sim 37\%$ oracle ack rate) with $\Delta = +2.20 \pm 1.44$ pp (non-inferiority $p < 0.001$; 5 seeds; Table 2)—a $\sim 99\times$ gap in suppression rate that underscores the need for pattern-level features beyond length. The near-zero length-only suppress rate confirms that Llama’s acknowledgments rarely fall below the character threshold; the near-zero accuracy delta (-0.10 pp) reflects the vacuity of the filter rather than any safety property, since 99.2% of messages pass through unchanged. This pattern reflects a fundamental difference in how backbones distribute communicative content across message lengths: Qwen concentrates acknowledgments in a narrow, low-token surface band amenable to a single threshold, whereas Llama embeds functionally equivalent content within verbose, lexically diverse outputs (65.6% bigram novelty; Appendix T), requiring pattern-level features for safe identification. The asymmetry implies that effective pruning detectors must be calibrated to backbone-specific surface morphology; no single-threshold mechanism can serve as a universal pruning rule across architectures.

Suggestive positive trend under heuristic pruning. Across both oracle and heuristic pruning on Llama/GSM8K, point estimates are consistently non-negative (oracle 10-seed $\Delta = +2.1 \pm 2.4$ pp, 9/10 positive; heuristic 5-seed $\Delta = +2.20 \pm 1.44$ pp, 5/5 positive, 95% CI $[+0.41, +3.99]$ pp), suggesting that removing redundant acknowledgments may reduce downstream reasoning drift—the tendency for agents to echo or elaborate on checker confirmations rather than advancing the solution. The heuristic’s over-suppression ($\sim 75\%$ vs. $\sim 37\%$ oracle ack rate) implies that $\sim 38\%$ of suppressed messages are non-acknowledgments; the positive accuracy delta despite this over-suppression is consistent with Llama’s one-directional error channel (0% correction, 14.3% injection; Appendix S.17)—removing error-injecting messages and their cascading effects outweighs any cost of suppressing a minority of informative messages. However, the current sample size is insufficient for causal inference on effect direction: non-inferiority is established at $\delta = -3$ pp ($t = 8.07$, $p < 0.001$); a separate one-sided test for positive effect yields $t = 3.42$, $p = 0.013$, $df = 4$, but a positive treatment effect would require a separate superiority test with substantially more power. We flag this as an exploratory finding warranting future investigation with larger N and controlled manipulation of ack content.

1651 S.23 Heuristic Threshold Sensitivity

Threshold sensitivity analysis. We vary the heuristic detector’s threshold parameters to assess whether pruning safety depends on specific threshold choices. Table 36 reports pruned accuracy across four detector variants on Llama-3.1-8B-Instruct / GSM8K ($N = 200$, seed = 42, greedy decoding, vLLM inference).

Pruned accuracy is stable across tested configurations (81.5–82.5%, ≤ 1.0 pp range) despite substantial variation in suppression intensity (23.1–75.2%). The keyword-only variant suppresses only 23.1% of messages yet achieves the same pruned accuracy as the more aggressive length-based variants,

Table 36: Heuristic threshold sensitivity (Llama/GSM8K, $N = 200$, single seed). Pruned accuracy is stable across detector configurations (81.5–82.5%). Baseline variability (77.7–81.5%) reflects vLLM inference non-determinism for V1–V3 (± 2 pp); V4’s lower baseline (77.7%) additionally reflects sample bias from 35% data loss.[†]

Variant	Configuration	Pruned Acc (%)	Suppress (%)	Baseline (%)
char<200	LEN=200, default kw+ngram	82.5	75.2	81.5
char<400	LEN=400, default kw+ngram	82.0	76.2	80.0
keyword-only	LEN=0, OVERLAP=999	82.5	23.1	79.5
ngram+length	KW=[], ngram+length only	81.5 [‡]	73.4	77.7 [‡]

[†]Each variant uses a separate baseline run; V1–V3 baseline differences (± 2 pp) are consistent with vLLM inference non-determinism observed across independent runs under identical configurations (cf. multi-run variance analysis above).
[‡]130/200 tasks completed; 70 tasks lost to vLLM infrastructure failure, introducing sample bias beyond inference non-determinism.

1659 confirming that the heuristic’s safety margin is not an artifact of a specific length threshold. This is a
1660 single-seed exploratory analysis; see Section 5.2 for multi-seed evidence under the default heuristic
1661 configuration.

1662 T Information-Theoretic Analysis of Acknowledgments

1663 To quantify why acknowledgment pruning under greedy decoding does not significantly degrade
1664 accuracy, we measure five information-theoretic properties of each message relative to its predecessor,
1665 grouped by communication function type. Metrics are computed on Qwen and Llama round-robin
1666 traces (300 tasks each); Gemma is excluded due to insufficient acknowledgment count ($n = 11$).

Table 37: Information-theoretic properties by function type (Qwen/GSM8K/RR, $N = 1,500$ messages). All ack vs. non-ack comparisons significant at $p < 10^{-10}$ (Mann–Whitney U) except where noted.

Metric	Ack	Info Prov.	Reasoning	Verif.	Coord.
Bigram novelty	0.057	0.317	0.749	0.434	—
Token length (median)	4	95	117	53	—
Jaccard overlap	0.686	0.653	0.313	0.482	—
Semantic similarity	0.783	0.896	0.823	0.841	—
Information density	0.971	0.565	0.566	0.685	—

Table 38: Information-theoretic properties by function type (Llama/GSM8K/RR, $N = 1,470$ messages).

Metric	Ack	Info Prov.	Reasoning	Coord.	Verif.
Bigram novelty	0.656	0.759	0.723	0.925	0.625
Token length (mean)	59.8	120.1	124.8	72.4	51.5
Jaccard overlap	0.342	0.332	0.336	0.191	0.283
Semantic similarity	0.636	0.817	0.763	0.453	0.706
Information density	0.774	0.472	0.519	0.741	0.718

1667 **Surface-level information content.** Table 39 complements the above metrics with lexical diver-
1668 sity (type–token ratio, TTR) and bigram/trigram overlap with the immediately preceding message,
1669 computed over word-level tokens. Across both backbones, acknowledgments exhibit the highest
1670 n-gram overlap with their predecessors (Qwen: 0.757 bigram, 0.733 trigram; Llama: 0.426, 0.363),
1671 confirming that they primarily echo prior content. Conversely, information provision and coordination
1672 messages show near-zero overlap (< 0.11 for Llama; < 0.26 for Qwen), indicating genuinely novel
1673 content. This quantitative gap directly explains the perturbation-sensitivity dissociation: perturbing
1674 an acknowledgment alters surface form but not functional information, since the replaced content
1675 was redundant with the preceding turn.

Table 39: Surface-level information metrics by function type and backbone (GSM8K/round-robin). *Bi-OL* and *Tri-OL*: fraction of current-message bigrams/trigrams also present in the preceding message. TTR: type-token ratio (unique words / total words). Gemma excluded ($n_{\text{ack}} = 11$).

Backbone	Function Type	N	Tokens	TTR	Bi-OL	Tri-OL
Qwen	Acknowledgment	951	39.2	0.820	0.757	0.733
	Verification	106	86.7	0.529	0.859	0.813
	Information provision	82	99.9	0.520	0.255	0.185
	Reasoning guidance	361	59.2	0.654	0.349	0.297
Llama	Acknowledgment	670	82.7	0.595	0.426	0.363
	Verification	112	115.8	0.422	0.420	0.369
	Information provision	192	62.1	0.665	0.235	0.184
	Reasoning guidance	376	69.2	0.624	0.168	0.105
	Coordination	120	37.4	0.845	0.105	0.064

Dual mechanism. Qwen acknowledgments are informationally vacuous: median bigram novelty is 0% and median Jaccard overlap with the preceding message is 1.0, meaning they are near-verbatim echoes. Pruning removes no unique information. Llama acknowledgments are longer and carry more novel content (bigram novelty 65.6%), but their pruning benefit arises from a different mechanism: they inject errors at 14.3% with 0% correction, so removal cuts an error-propagation channel rather than removing redundancy.

Per-round acknowledgment trends. The proportion of acknowledgments varies across conversation rounds and backbones. Qwen shows monotonic increase from 0% (round 1) to 96% (round 5), consistent with later rounds serving primarily confirmatory roles. Llama exhibits an inverted-U pattern peaking at 81% in round 3 before declining, suggesting a transition from acknowledgment to substantive re-engagement in later rounds. Gemma produces near-zero acknowledgments across all rounds, consistent with its position-organized profile where the acknowledgment category is functionally absent.

T.1 Heuristic Detector Performance

Backbone-specific heuristic detectors using only deterministic features (length, keywords, n-gram repetition) achieve F1 = 0.845 (Qwen, rule: character length < 30) and 0.805 (Llama, composite rule: keyword matching + length < 300 + 4-gram repetition > 0.3) with 0–1 pp pruning gap versus oracle labels, validated via 5-fold GroupKFold CV (grouped by task). Composite rules generalize cross-domain: Llama F1 > 0.71 on StratQA and ARC. Qwen’s single-threshold rule degrades cross-domain (F1 = 0.21 on StratQA), motivating the multi-feature approach for deployment. Full development details appear in §4.6.

U Scale Validation: Qwen2.5-14B

We extend the scale trajectory analysis (Section 5) with detailed results for Qwen2.5-14B-Instruct on GSM8K under round-robin topology, bridging the 7B and 72B endpoints.

U.1 Function Annotation

GPT-4o annotation of 600 agent messages yields an acknowledgment rate of 31.7% (190/600), closely matching the 72B relabeling rate (29.7%). The full function distribution is: VERIFICATION 191, ACKNOWLEDGMENT 190, REASONING_GUIDANCE 162, INFORMATION_PROVISION 57, COORDINATION 0. The near-identical acknowledgment proportions across 14B and 72B suggest that communication function distributions are stable within the Qwen2.5 family at these scales.

Cross-model annotation agreement. Qwen2.5-72B-Instruct-AWQ relabeling yields an acknowledgment rate of 29.7% (178/600), within 2 pp of the GPT-4o rate (31.7%), confirming annotator consistency at this model scale.

1709 U.2 Heuristic Detector

1710 A character-length threshold of 80 achieves $F1 = 0.837$ (precision = 0.799, recall = 0.879) against
1711 GPT-4o labels. The higher threshold compared to 7B (char_threshold = 30) reflects 14B’s tendency
1712 to produce longer acknowledgment messages. Despite the different threshold, classification quality
1713 remains comparable to the 7B detector ($F1 = 0.845$).

1714 U.3 Simulate Pruning

1715 Offline simulate pruning based on 72B annotations (which closely agree with GPT-4o) yields:
1716 baseline accuracy 95% (from offline traces), ack-only accuracy 95%, $\Delta = 0$ pp, with 18.3% token
1717 savings. The zero delta is consistent with acknowledgment content carrying no marginal task-relevant
1718 information at 14B scale, matching the 7B finding.

1719 U.4 Online Pruning

1720 Online pruning (skip mode, char_threshold = 80, $N = 100$) yields: baseline 90%, ack-only 93%,
1721 $\Delta = +3$ pp, with 5.8% suppression rate (29/500 messages). The low suppression rate—substantially
1722 below the annotated acknowledgment proportion (31.7%)—reflects the conservative char_threshold:
1723 at threshold = 80, only the shortest acknowledgments are suppressed online, while many genuine
1724 acknowledgments exceed 80 characters and pass through.

1725 **Unified threshold validation.** To enable fair cross-scale comparison, we re-run online pruning
1726 with char_threshold = 30 (matching the 7B and 72B threshold). This yields: baseline 91%, ack-only
1727 95%, $\Delta = +4$ pp, with 6.4% suppression rate (32/500 messages). The minimal increase from
1728 5.8% to 6.4% indicates that the low suppression rate is intrinsic to 14B’s acknowledgment length
1729 distribution—14B generates few short (< 30 character) acknowledgments—rather than an artifact of
1730 threshold calibration. The heuristic thus captures only $\sim 20\%$ of true acknowledgments at 14B (6.4%
1731 suppressed vs. 31.7% true ack rate per GPT-4o), illustrating that length-based heuristic transferability
1732 varies by scale. This makes the 14B online result a test of partial heuristic coverage rather than
1733 a full pruning safety test; the simulate pruning result ($\Delta = 0$ pp with oracle labels) provides the
1734 complementary full-removal evidence.

1735 **Interpreting the low suppression rate.** Both thresholds yield $< 7\%$ suppression, meaning the
1736 delta is based on removing only ~ 30 messages out of 500. While the direction is consistently positive
1737 (no degradation), the small number of suppressed messages limits the strength of inference about
1738 what would happen under more aggressive pruning. The simulate pruning result ($\Delta = 0$ pp at full
1739 acknowledgment removal) provides complementary evidence that complete removal is safe.

1740 **Run-to-run variance.** Under greedy decoding, single-run point estimates carry ± 1 – 2 pp residual
1741 variance from GPU numerical precision (cf. ten-run replication for 7B: ± 0.99 pp). The $+3$ pp and
1742 $+4$ pp deltas are within this variance band and should not be interpreted as genuine improvements.

1743 U.5 Full-Scale 72B Preliminary Probe ($N = 265$)

1744 The exploratory 72B probe ($N = 100$, $\Delta = 0$ pp) uses a sample too small for powered inference. We
1745 extend to $N = 265$ with oracle (GPT-4o 5-class) labels to provide a preliminary single-seed scale
1746 test.

1747 **Online oracle pruning.** Qwen2.5-72B-Instruct-AWQ on GSM8K ($N = 265$): baseline 92.1%,
1748 pruned 92.1%, $\Delta = +0.0$ pp ($p = 1.000$), with 66.8% message suppression. The suppression rate at
1749 72B (66.8%) can be compared to 7B (56.3%) to assess whether larger models produce proportionally
1750 different acknowledgment rates.

1751 **Cross-scale comparison.** Table 40 summarizes oracle pruning deltas across the Qwen2.5 scale
1752 trajectory.

Table 40: Pruning Δ (pp) across Qwen2.5 scales on GSM8K (oracle where available, heuristic otherwise; greedy decoding).

Scale	N	Baseline	Pruned	Δ	Supp.
7B	500	88.8	89.6	+0.8	56.3%
14B [†]	100	90/91	93/95	+3/+4	5.8/6.4%
72B	100	—	—	0	—
72B	265	92.1	92.1	+0.0	66.8%

[†] 14B uses char-length heuristic (thresholds 80/30); oracle online pruning was not run at this scale. Offline oracle ($\Delta = 0$ pp) provides complementary full-removal evidence.

U.6 MAS Deployment Positioning

Our study analyzes communication *within* multi-agent systems, not whether multi-agent architectures outperform single-agent alternatives—a question addressed by existing MAS benchmarks Du et al. [2025a], Zhu et al. [2025]. The taxonomy is an empirically-derived analytical framework for understanding inter-agent message functions; its value lies in identifying which messages are safe to remove from an existing MAS pipeline, not in justifying MAS over single-agent approaches. Single-agent baselines are orthogonal: they would test whether agents should communicate at all, whereas we test which communications are redundant given that agents *do* communicate. This distinction parallels neural pruning LeCun et al. [1990], where the question is which weights to remove from a trained network, not whether the architecture should have fewer layers.

U.7 Scale Trajectory of Pruning Safety

Ack-pruning direction is directionally consistent across Qwen2.5 7B and 14B scales, with preliminary single-seed evidence at 72B: 7B ($\Delta = +0.8$ pp, $N = 500$), 14B (offline oracle $\Delta = 0$ pp), and 72B ($N = 265$, $\Delta = +0.0$ pp, zero discordant flips, $p = 1.000$, 66.8% suppression; Table 2; Appendix S.3). No significant degradation is observed at any individual scale point; the 72B result is preliminary evidence (single seed, insufficient to estimate seed-to-seed variance) with $N = 265$ (truncated from 500 due to API rate limits; binomial 95% CI upper bound on flip rate $\approx 1.1\%$). The 14B result is limited by low heuristic suppression (6.4%); the offline oracle ($\Delta = 0$ pp) provides complementary evidence. Individual-message perturbation at 72B remains unmeasured; the dissociation is inferred from 7B perturbation data.

V Cross-Experiment Summary

Table 41 consolidates all oracle and pruning experiments across backbones, scales, domains, and temperatures. Cohen’s h quantifies effect size on the arcsine-difference scale; all non-boundary $|h| < 0.20$, falling in the “negligible” range by conventional thresholds [Cohen, 1988]. For the 72B zero-flip result, $h = 0$ exactly—the strongest possible “no effect” evidence.

W Efficiency Analysis

Computational efficiency. The primary efficiency gain from acknowledgment pruning is *inference call reduction*: each suppressed message eliminates one full LLM generation call. Oracle-guided pruning suppresses **56.0%** of inter-agent messages on Qwen2.5-7B (aggregated over ten seeds, $N=2,000$; **2.80 per task**) and **36.5%** on Llama-3.1-8B (ten seeds, $N=1,939$; **1.84 per task**), translating directly to proportional wall-time savings under roughly uniform per-round latency (Table 42). Token savings are substantially lower—approximately **10.1%** on Qwen and **27.5%** on Llama—because acknowledgments are much shorter than substantive messages (Qwen ack median: 4 tokens vs. verification/reasoning: 53–117 tokens). The disparity is itself informative: *ack pruning’s value lies in eliminating redundant inference calls, not in reducing token volume*. In deployment, backbone-specific heuristic detectors (F1 = 0.845 for Qwen, 0.805 for Llama; §4.6) achieve these savings at zero marginal cost, requiring no LLM annotation.

Table 41: Full cross-experiment pruning matrix. Δ : pruned – baseline accuracy (pp). p : McNemar χ^2 with continuity correction (single-run) or paired t (multi-seed); exact binomial p -values agree within 0.001 for all entries. TOST: equivalence p at stated margin. h : Cohen’s arcsine effect size. Non-boundary $|h| < 0.20$ (negligible); Qwen3-8B reaches “small” ($|h| = 0.38$); MATH at $N=200$ is negligible ($|h| = 0.048$). [†]Fisher combined across two independent $N=500$ runs. [‡]Heuristic detector, not oracle. [§]Simulate pruning (offline with oracle labels). ^{||} Single-seed; TOST requires multi-seed variance estimate.

Backbone	Domain	Temp	N	Seeds	Protocol	Δ (pp)	p	TOST	h
<i>Oracle pruning (GPT-4o 5-class labels)</i>									
Qwen-7B	GSM8K	0	500	1	Oracle	+0.8	.343	—	+ .026
Qwen-7B	GSM8K	0.7	500	1	Oracle	+0.6	.710	—	+ .018
Qwen-7B	StratQA	0	500	1	Oracle	+0.2	1.000	—	+ .005
Llama-8B	GSM8K	0	500	1	Oracle	−0.6	.710	—	− .016
Llama-8B	GSM8K	0.7	1,000	2 [†]	Oracle	−2.2	.291	—	− .054
Llama-8B	StratQA	0	238	1	Oracle	+5.0	.180	—	+ .111
Qwen-72B	GSM8K	0	265	1	Oracle	+0.0	1.000	—	.000
<i>Multi-seed replication</i>									
Qwen-7B	GSM8K	0	200×10	10	Oracle	+0.40	—	±1 pp: .048*	+ .013
Qwen-7B	GSM8K	0	200×10	10	Heuristic [‡]	−0.05	.876	±1 pp: .007**	− .002
Llama-8B	GSM8K	0	~194×10	10	Oracle	+2.08	.023*	±3 pp: <.001***	+ .057
Llama-8B	StratQA	0	200×8	8	Heuristic [‡]	+2.81	.084	±3 pp: .002**	+ .060
<i>Temperature robustness (heuristic detector)</i>									
Qwen-7B	GSM8K	0.7	500	1	Heuristic [‡]	−0.6	.728	—	− .019
Llama-8B	GSM8K	0.7	500	1	Heuristic [‡]	−3.4	.100	—	− .084
Gemma-9B	GSM8K	0.7	100	1	Offline	+3.0	.453	—	+ .076
<i>Scale validation (Qwen2.5 family)</i>									
Qwen-14B	GSM8K	0	100	1	Simulate [§]	0.0	—	—	.000
Qwen-72B	GSM8K	0	100	1	Heuristic [‡]	0.0	—	—	.000
<i>Cross-domain offline (ack-only vs. baseline)</i>									
Qwen-7B	ARC	0	250	1	Offline	+0.8	—	—	+ .019
Llama-8B	ARC	0	250	1	Simulate [§]	+4.0	.013*	—	+ .088
Gemma-9B	GSM8K	0	96	1	Offline	−1.0	—	—	− .023
<i>Online heuristic pruning (from-scratch)</i>									
Llama-8B	GSM8K	0	1,000	1	Online	+2.1	.004**	—	+ .055
Qwen-7B	StratQA	0	200×5	5	Heuristic [‡]	+0.70	.160	±3 pp: <.001**	+ .015
<i>Random baseline</i>									
Qwen-7B	GSM8K	0	500	1	Token-matched	+0.8	.343	—	+ .026
<i>Topology / agent count</i>									
Qwen-7B (3-agent)	GSM8K	0	100	1	Online	−1.0	1.000	—	− .036
<i>Boundary conditions (expected degradation)</i>									
Qwen3-8B	GSM8K	0	100	1	Heuristic [‡] (char<80)	−14.0	<.03*	—	− .380
Qwen-7B	MATH	0	200 (102 v)	1	Online	−2.0	.625	—	− .048

Oracle cost and upper-bound positioning. The GPT-4o oracle classifier used for pruning validation incurs **\$13.90–\$14.42 per 1,000 tasks** (3,475–3,605 classification tokens per task at current API pricing), with per-message classification adding latency overhead. This positions oracle pruning as a *research upper bound*: it isolates true acknowledgments with high precision, enabling controlled measurement of pruning safety (§5.2), but its cost and latency make it impractical for deployment. The heuristic detectors (§4.6) replace the oracle at zero cost—simple deterministic rules (Qwen: character length < 30; Llama: keyword + length + repetition) executable in microseconds, with no API dependency. For open-weight models on local infrastructure—the setting of this paper—savings manifest as reduced GPU-seconds and lower round-trip latency; for API-served models, each suppressed message eliminates a full API call whose cost includes per-call overhead beyond raw token pricing.

Table 42: Efficiency of oracle-guided ack pruning on GSM8K. Calls saved = suppression rate; token savings are lower because acks are short (median 4 tokens). Oracle cost is for GPT-4o 5-class classification; heuristic detectors achieve comparable savings at zero cost (\$4.6).

Backbone	N	Calls Saved	Msgs / Task	Est. Token Saved	Oracle / Task	Oracle Cost / 1K
Qwen2.5-7B	2,000	56.0%	2.80 / 5.0	$\sim 10.1\%$	3,475 tok	\$13.90
Llama-3.1-8B	1,939	36.5%	1.84 / 5.0	$\sim 27.5\%$	3,605 tok	\$14.42

X Exploratory Multi-Agent and Alternative Topology Pilots

To probe whether acknowledgment patterns generalize beyond two-agent solver-checker configurations, we conduct two exploratory pilots using Qwen2.5-7B-Instruct on GSM8K and StrategyQA ($N = 100$ each, greedy decoding).

Three-agent solver-checker-critic. We extend the standard two-agent pipeline with a third agent (LogicCritic) that reviews the checker’s verification for completeness, cycling in round-robin order over 7 rounds. The heuristic detector ($\text{char} < 30$) suppresses 37% of agent messages (222/600), with suppression concentrated in checker and critic roles (Table 43). Pruned accuracy (91%) is within 1 pp of baseline (92%; McNemar $p = 1.0$, 1 discordant pair). The ack distribution across agent positions—19% (solver), 40% (checker), 41% (critic)—suggests that downstream agents produce more acknowledgments, consistent with the cascade dynamics observed in two-agent systems.

Debate (advocate–challenger). We replace the solver-checker topology with symmetric argumentation: an Advocate argues for an answer while a Challenger identifies weaknesses, cycling over 6 rounds on StrategyQA (yes/no). The heuristic detector suppresses only 0.21% of messages (1/485), with 99 of 100 tasks producing zero suppressed messages. Baseline accuracy is 62% (pruned 63%, $\Delta = +1$ pp). The near-zero suppression rate indicates that debate agents produce long argumentative responses rather than short acknowledgment-like messages, suggesting that ack prevalence is topology-dependent rather than a universal property of multi-agent LLM communication. An alternative explanation is that the heuristic ($\text{char} < 30$) is poorly calibrated for debate responses; oracle annotation was not performed for this pilot.

Table 43: Exploratory topology pilots (Qwen2.5-7B, $N = 100$, greedy decoding). Suppression uses the heuristic detector ($\text{char} < 30$); no oracle annotation was performed for the debate pilot.

Topology	Domain	Agents	Baseline	Pruned	Δ (pp)	Suppress %
3-Agent (S-C-Cr)	GSM8K	3	92%	91%	−1.0	37.0%
Debate (A-Ch)	StrategyQA	2	62%	63%	+1.0	0.21%

Both pilots are exploratory ($N = 100$, single backbone, heuristic detection only). Definitive claims about multi-agent or alternative-topology pruning require larger-scale validation with oracle annotation.

Y Artifact Description

All code, data, and annotation materials are released under the MIT License at [anonymized].

Y.1 Released Artifacts

Code repository. The repository contains the full experimental pipeline:

- **Trace collection** (`src/trace_collection.py`): multi-agent conversation recording via AG2, supporting round-robin, sequential, and selector topologies.
- **Functional annotation** (`src/functional_annotation.py`, `src/gpt4o_annotation.py`): LLM-based 5-class message function annotation with cross-model agreement.

- **Acknowledgment detection** (src/heuristic_ack_detector.py, scripts/train_ack_classifier.py): rule-based and trained neural classifiers for identifying acknowledgment messages.
- **Perturbation analysis** (src/local_impact.py): per-message remove and neutralize ablations with LDIM computation.
- **Pruning validation** (src/pruning_validation.py): offline simulation and online pruning with oracle or heuristic detection.
- **Statistical analysis** (src/analysis.py): McNemar tests, TOST equivalence tests, likelihood-ratio tests, and figure generation.

YAML configuration files in configs/ specify all model, agent, data, and topology parameters for reproducibility.

Datasets.

- **Sample traces** (data/sample_traces/): representative annotated traces from one backbone-domain configuration, with complete traces for all configurations available in the repository.
- **Human annotation data** (data/human_annotation/): annotation sheets from three independent annotators in CSV format, with inter-annotator agreement scripts (Cohen’s κ , Fleiss’ κ , Krippendorff’s α).
- **Classifier training data** (data/classifier/): train/validation/test splits for acknowledgment classifier training.

Annotation materials. The repository includes the annotation guideline document (docs/annotation_guide.md) and a standalone web-based annotation interface (data/human_annotation/annotator_standalone.html) used in the human validation study described in Section 4.

Y.2 Protocol Reuse

Researchers can apply our analysis framework to their own multi-agent systems in three steps:

1. **Collect multi-agent traces.** Record inter-agent message logs from the target system. The pipeline accepts JSONL traces; src/mast_utils.py provides utilities for converting from the MAST trace format. For AG2-based systems, src/trace_collection.py handles trace recording directly.
2. **Annotate communication functions.** Run src/functional_annotation.py to classify each message into the five-type taxonomy (Acknowledgment, Information Provision, Reasoning Guidance, Verification, Coordination). Alternatively, use the heuristic detector (src/heuristic_ack_detector.py) for lightweight acknowledgment-only classification without an external LLM API.
3. **Evaluate pruning trade-offs.** Run src/pruning_validation.py with --mode online_pruning to measure the accuracy–efficiency trade-off of removing detected acknowledgment messages in live conversations. The statistical analysis script (src/analysis.py) produces McNemar tests, equivalence tests, and summary figures.

Y.3 License and Ethics

The code and annotation data are released under the MIT License. The underlying benchmark datasets (GSM8K, ARC-Challenge, StrategyQA) retain their original licenses. All three model backbones use permissive open-weight licenses. No personally identifiable information is contained in any released artifact; all data consists of LLM-generated text and researcher annotations thereof.

NeurIPS Paper Checklist

1. **Claims Yes:** All claims are supported by empirical evidence from our experiments. Ablation methodology is described as “perturbation-based” to accurately reflect its observational nature.

- 1881 2. **Limitations Yes:** Section 6 discusses limitations including model scale (7B primary, sub-
1882 stantive 14B oracle, 72B oracle $N=265$), topology coverage (two-agent with exploratory
1883 three-agent pilot), LLM annotation scope (κ variation), and greedy decoding sensitivity.
- 1884 3. **Theory Assumptions and Proofs N/A:** Our work is empirical; no theoretical proofs are
1885 presented.
- 1886 4. **Experimental Result Reproducibility Yes:** Section 4 specifies all models, hyperparam-
1887 eters, framework versions, and statistical tests. An anonymous code repository containing
1888 the complete experimental pipeline, configuration files, and sample data is provided at
1889 [anonymized]. GSM8K, ARC-Challenge, and StrategyQA are publicly available bench-
1890 mark datasets.
- 1891 5. **Open access to data and code Yes:** Code, traces, and annotations are provided in an
1892 anonymous repository at [anonymized] under the MIT License. GSM8K, ARC-Challenge,
1893 and StrategyQA are publicly available benchmark datasets.
- 1894 6. **Experimental Setting/Details Yes:** Section 4 provides full experimental details including
1895 model names, parameter counts, generation settings, annotation protocol, and statistical
1896 framework.
- 1897 7. **Experiment Statistical Significance Yes:** All experiments report exact p -values from
1898 McNemar exact tests, TOST equivalence tests, likelihood-ratio tests (LRT), and Fisher exact
1899 tests. The primary test family (9 LRT tests) receives Benjamini–Hochberg correction at
1900 $\alpha = 0.05$. Effect sizes, confidence intervals, minimum detectable effects, and statistical
1901 power are reported for all key comparisons. Ten-run online replication provides variance
1902 estimates and formal equivalence (Qwen $\Delta = -0.05 \pm 0.99$ pp, TOST $p = 0.007$ at ± 1 pp;
1903 95% CI $[-0.75, +0.65]$ pp). Full statistical details appear in Appendix A.
- 1904 8. **Compute Resources Yes:** Experiments were conducted on $4 \times$ NVIDIA RTX PRO 6000
1905 Blackwell Server Edition GPUs (96 GB each). Total compute including all replications,
1906 oracle validation, and ablations is approximately 120 GPU-hours.
- 1907 9. **Code Of Ethics Yes:** This research studies existing open-weight models on a public
1908 benchmark. No new models are trained. Human annotation involved four research team
1909 members classifying LLM-generated messages; no external participants or sensitive data
1910 were involved.
- 1911 10. **Broader Impacts Yes:** Our work characterizes communication in two-agent LLM systems
1912 to improve efficiency and interpretability. Our finding that structurally minimal messages
1913 (acknowledgments) are disproportionately impactful implies a potential adversarial vector:
1914 an attacker who can inject or suppress acknowledgment-like messages in a two-agent pipeline
1915 could cause outsized behavioral disruption with minimal detectable content. Defenses should
1916 monitor structural patterns, not just content, for integrity. Beyond this, we do not foresee
1917 negative societal impacts beyond those inherent to LLM deployment generally.
- 1918 11. **Safeguards Yes:** The paper identifies acknowledgment injection as a potential adversarial
1919 vector (Section 6). This is a research finding, not a risk introduced by our methodology.
- 1920 12. **Licenses Yes:** GSM8K is MIT-licensed. All three model backbones use permissive open-
1921 weight licenses. AG2/AutoGen is Apache 2.0.
- 1922 13. **New Assets Yes:** We release interaction traces, function type annotations, and ablation results
1923 in the anonymous repository at [anonymized] under the MIT License. See Appendix Y
1924 for a complete artifact description.
- 1925 14. **Crowdsourcing and Research with Human Subjects Yes:** Research team members classi-
1926 fied 225 inter-agent messages into 5 communication function types for human validation of
1927 the LLM-derived taxonomy (Section 6). No external crowdsourcing was used. No personally
1928 identifiable or sensitive data were collected; annotators labeled only LLM-generated text.
- 1929 15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human
1930 Subjects N/A:** The human annotation task involved only research team members classifying
1931 LLM-generated text, with no collection of personal data. This does not constitute human
1932 subjects research under standard IRB definitions.
- 1933 16. **Declaration of LLM Usage Yes:** GPT-4o is used for cross-model annotation of com-
1934 munication function types (Section 4.1). Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct,

1935 and Gemma-2-9B-IT serve as both backbone models generating conversations and cross-
1936 annotators. The annotation protocol, prompts, and inter-annotator agreement are documented
1937 in Section 4.1 and Appendices D, F. Human validation on 225 messages confirms taxonomy
1938 reliability ($\kappa = 0.76$). LLMs were not used for writing or editing the paper text.